

Application of Data Analytics with Tableau

(Part 1)



Outline:

PART 1:

- Getting Started
- Connecting to Data
- Overview of Tableau – Basics and Concepts
- Creating Basic Visualization Charts
- Organizing Data

PART 2:

- Analytics Tab
- Mapping the Data
- Data Blending
- Dashboard and Stories

Tableau Product

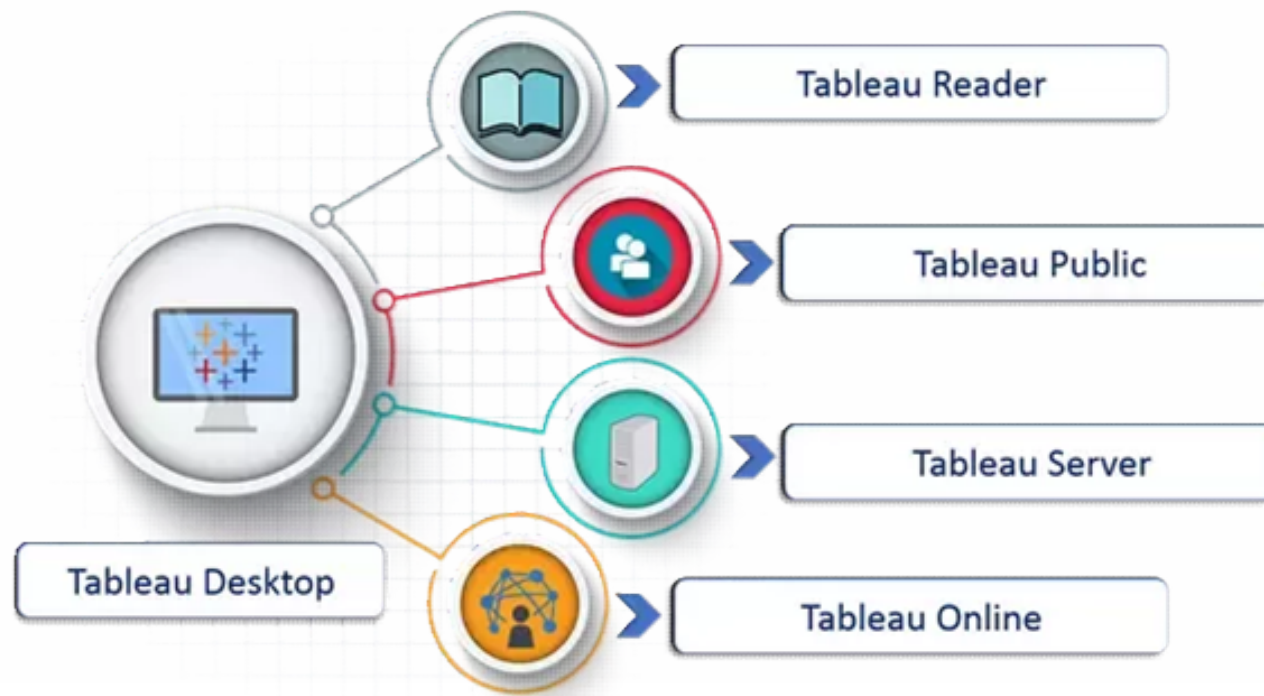


Tableau Product (cont'd)

Tableau Desktop	Tableau Desktop Public
Paid (free for 14 days)	Free
Supports all kinds of data connections including local files and databases.	Supports only local file connections.
Workbook file can be saved locally as *.twb or *.twbx format.	Workbook file cannot be saved locally, need to be saved and uploaded to Tableau Public Server
Privacy	No privacy
Can work offline	Must work online in order to save the analysis file

Tableau Product (cont'd)

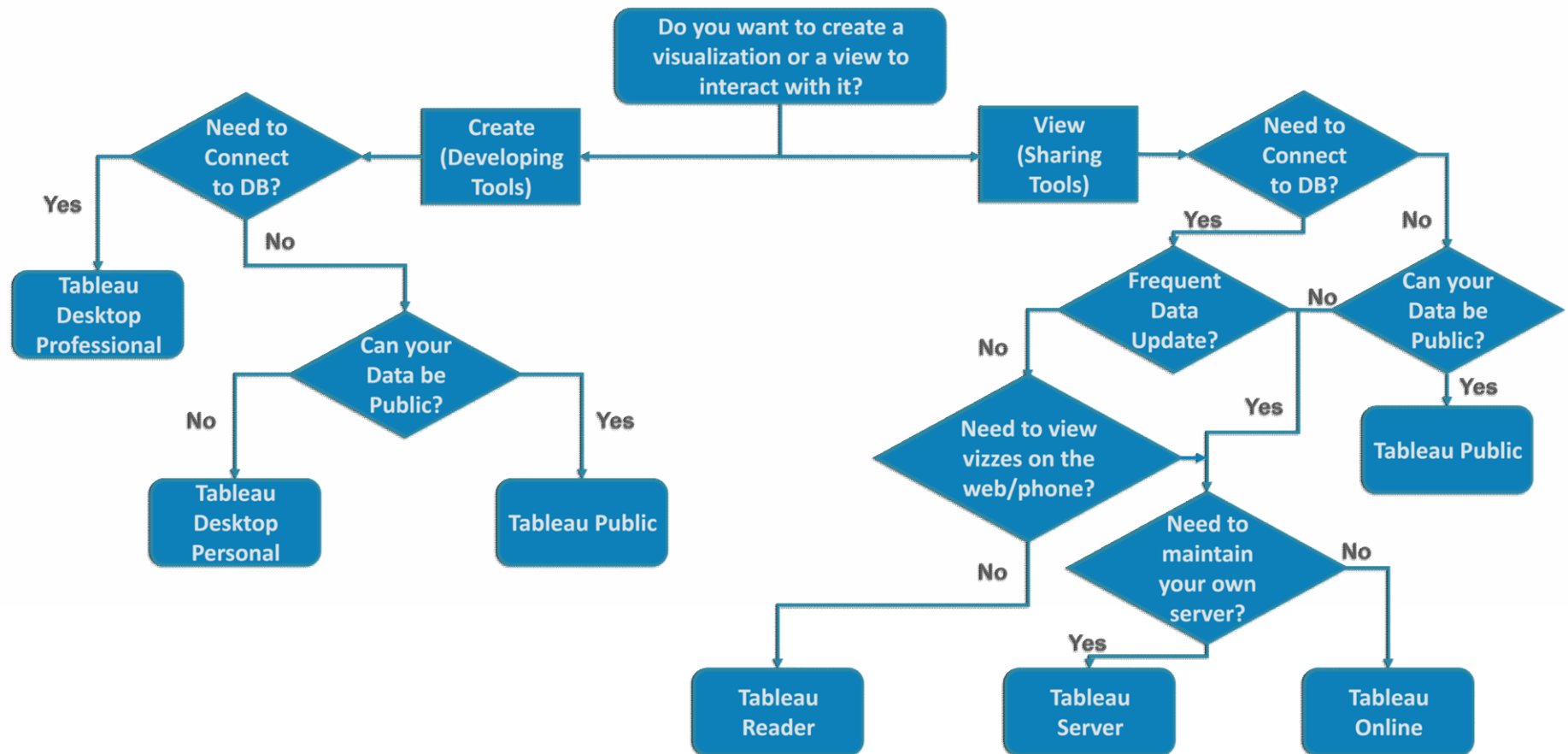
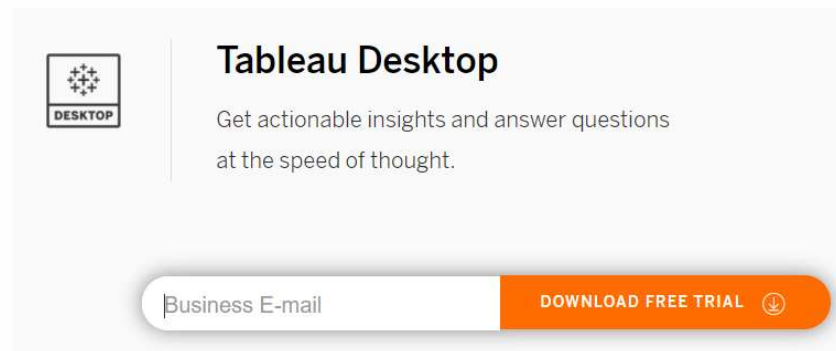


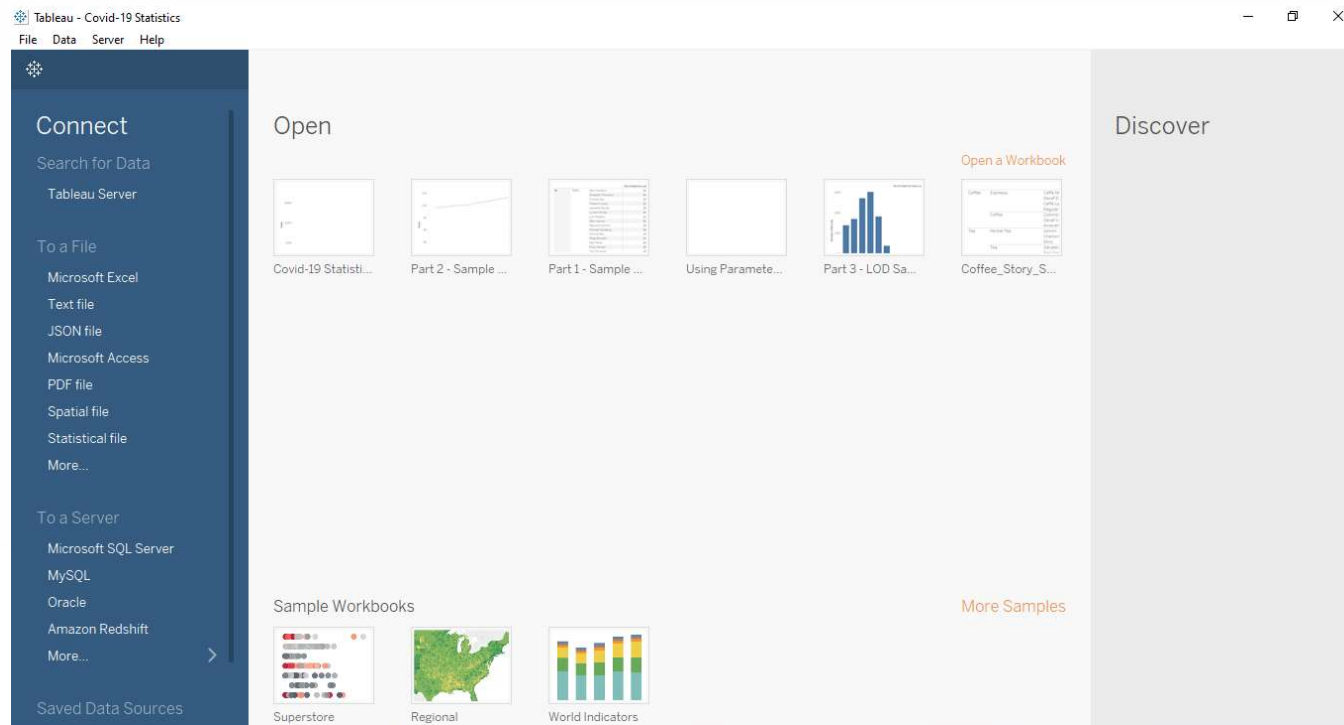
Tableau Install

- <https://www.tableau.com/products/trial>



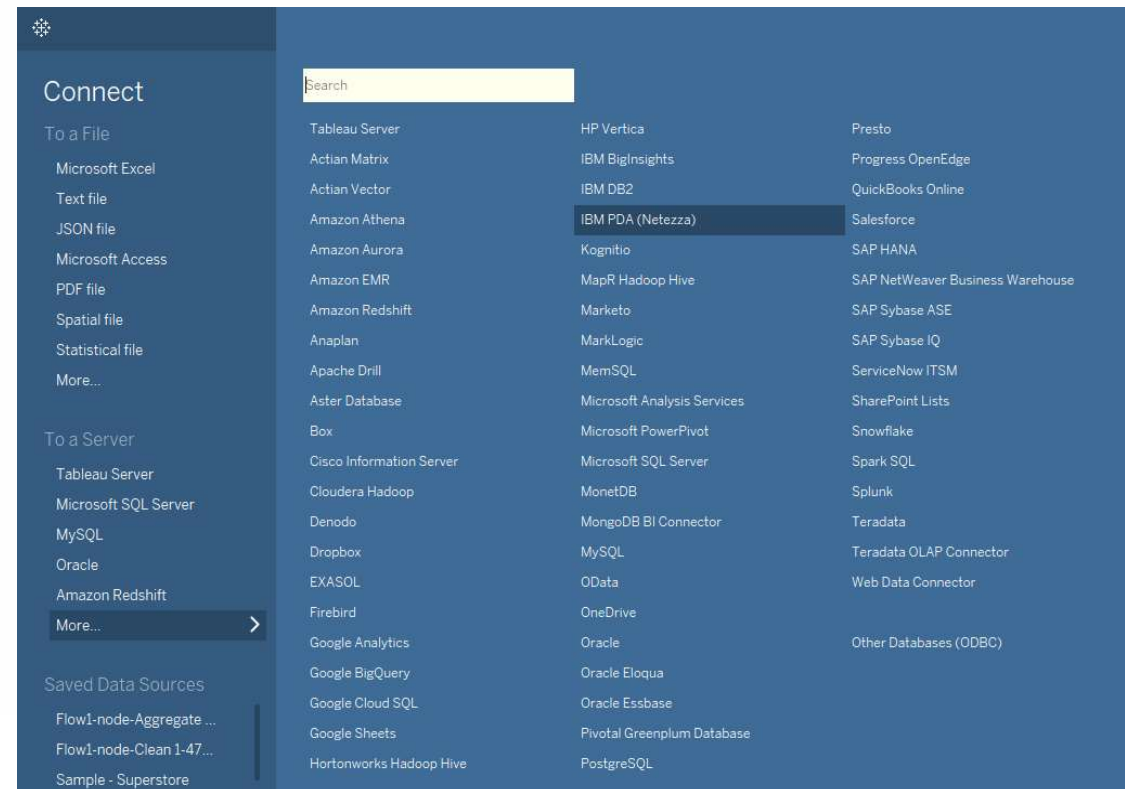
Tableau's Homepage

Open the Tableau by clicking on the Tableau icon  . You will see a Tableau's Homepage , where you have the option to choose your data connection or open the existing file.



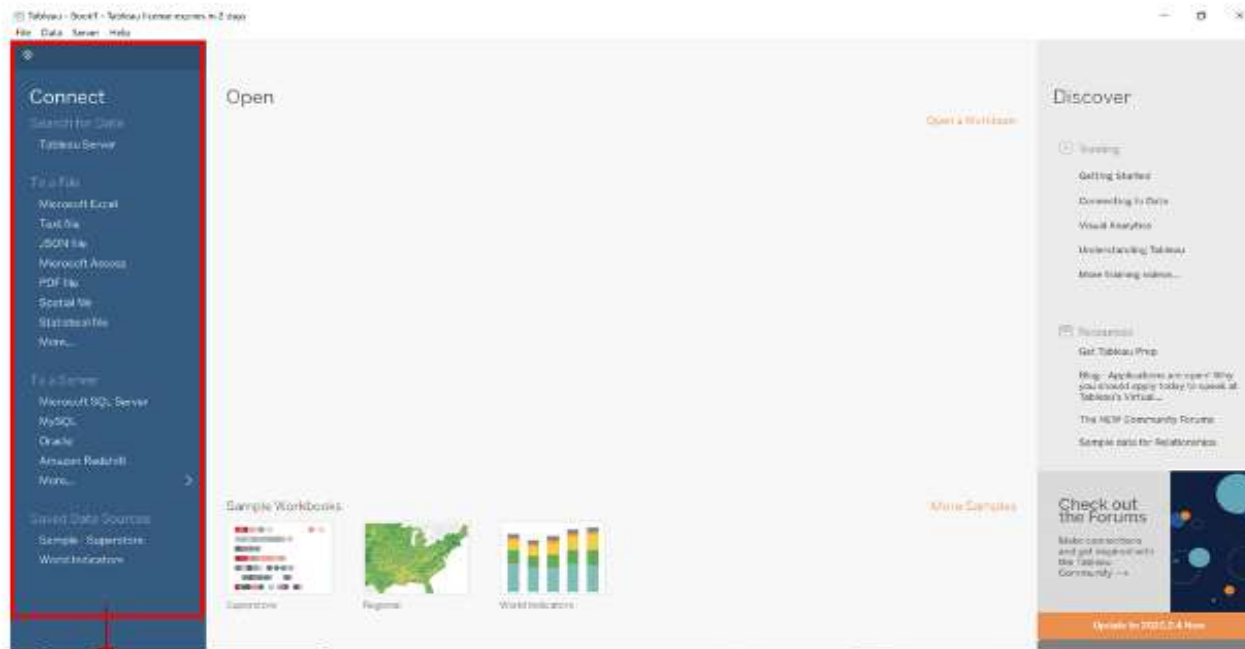
Connect To Data

- ❏ Tableau can connect to many data sources regardless of whether they're local files or databased on-premises or in the cloud.
- ❏ A workbook can have multiple data source.
- ❏ Joining Tables and Unions



Connect To Data (cont'd)

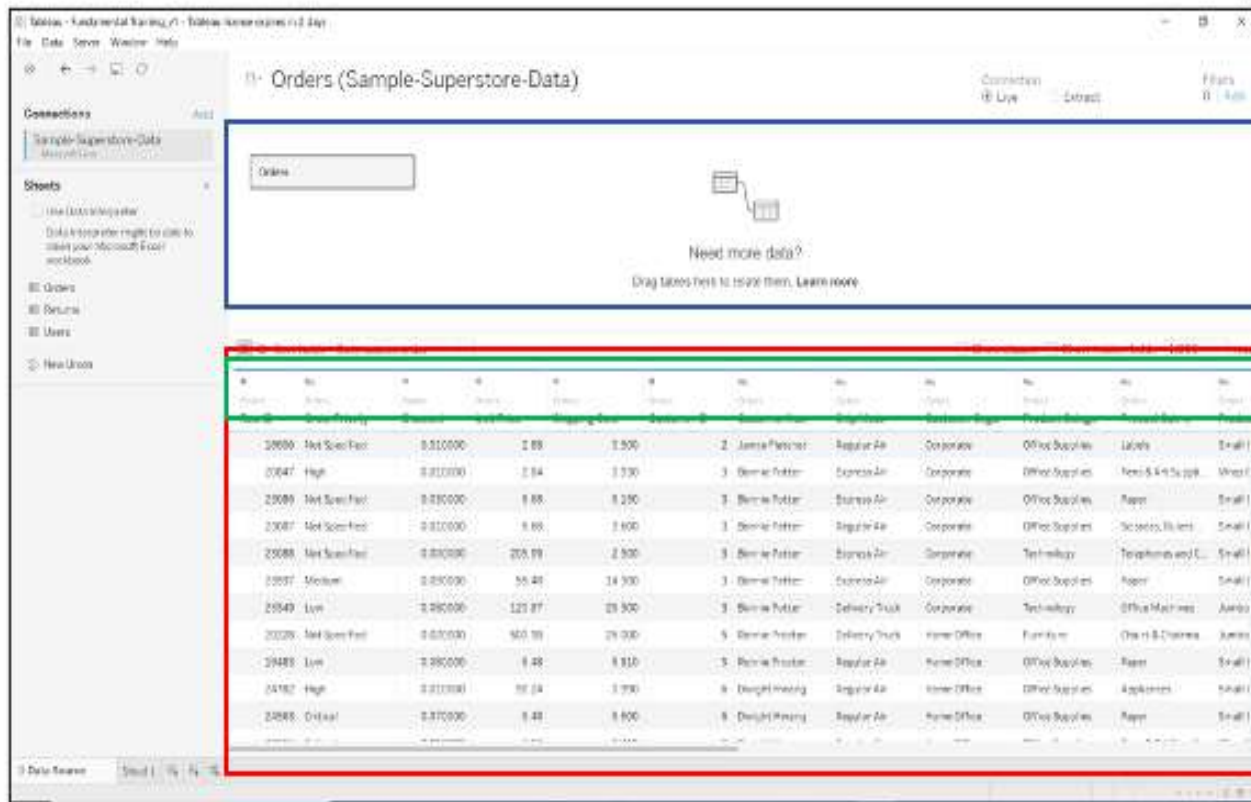
1. On the Connect Pane, click Microsoft Excel.
2. Navigate to the file on your machine (desktop/laptop) and open a file: **US APAC – Superstore Spreadsheet.xlsx**.



Connect Pane

Connect To Data (cont'd)

- Click and drag out the sheet named **Order** into the canvas (*Drag tables here*). Part of the data can be viewed on the preview pane.



The screenshot shows the 'Orders (Sample-Superstore-Data)' interface. The top section is the 'Canvas', which contains a 'Need more data?' message and a 'Drag tables here to view them. Learn more.' prompt. Below the canvas is the 'Data type info' section, which displays a table of data. The bottom section is the 'Preview pane', which shows a preview of the data from the 'Orders' table. The preview pane is highlighted with a red box and a red arrow pointing to it from the label 'Preview pane'.

Canvas

Data type info

Preview pane

Order ID	Product Name	Quantity	Price	Discount	Customer Name	Product Category	Product Subcategory	Product Brand	Product Size	Product Color
10660	Mid Size Pallet	0.010000	2.88	0.500	2. James Parker	Regular Air	Corporate	Office Supplies	Labels	Small
10647	High	0.010000	2.84	0.330	3. Dennis Parker	Express Air	Corporate	Office Supplies	Paper 5.4x5.5gk	Small
10689	Mid Size Pallet	0.010000	8.88	0.280	3. Dennis Parker	Express Air	Corporate	Office Supplies	Paper	Small
10687	Mid Size Pallet	0.010000	8.88	0.600	3. Dennis Parker	Regular Air	Corporate	Office Supplies	Scissors, Blades	Small
10688	Mid Size Pallet	0.010000	205.99	2.500	3. Dennis Parker	Express Air	Corporate	Technology	Telephones and C.	Small
10697	Medium	0.010000	55.48	14.300	3. Dennis Parker	Express Air	Corporate	Office Supplies	Paper	Small
10649	Low	0.010000	123.87	08.500	3. Dennis Parker	Delivery Truck	Corporate	Technology	Office Machines	Audio
10726	Mid Size Pallet	0.010000	60.99	29.000	5. Dennis Parker	Delivery Truck	Home Office	Furniture	Chair & Seating	Audio
10489	Low	0.010000	6.48	8.810	5. Dennis Parker	Regular Air	Home Office	Office Supplies	Paper	Small
10482	High	0.010000	88.04	0.790	6. Dwight Herring	Regular Air	Home Office	Office Supplies	Appliances	Audio
10486	Ordnance	0.010000	8.48	8.800	6. Dwight Herring	Regular Air	Home Office	Office Supplies	Paper	Small

Connect To Data (cont'd)

- Once Tableau has read the data, it will assign the data type to each field. If it wasn't correctly assigned, you can make changes on the **Data Sources** page.

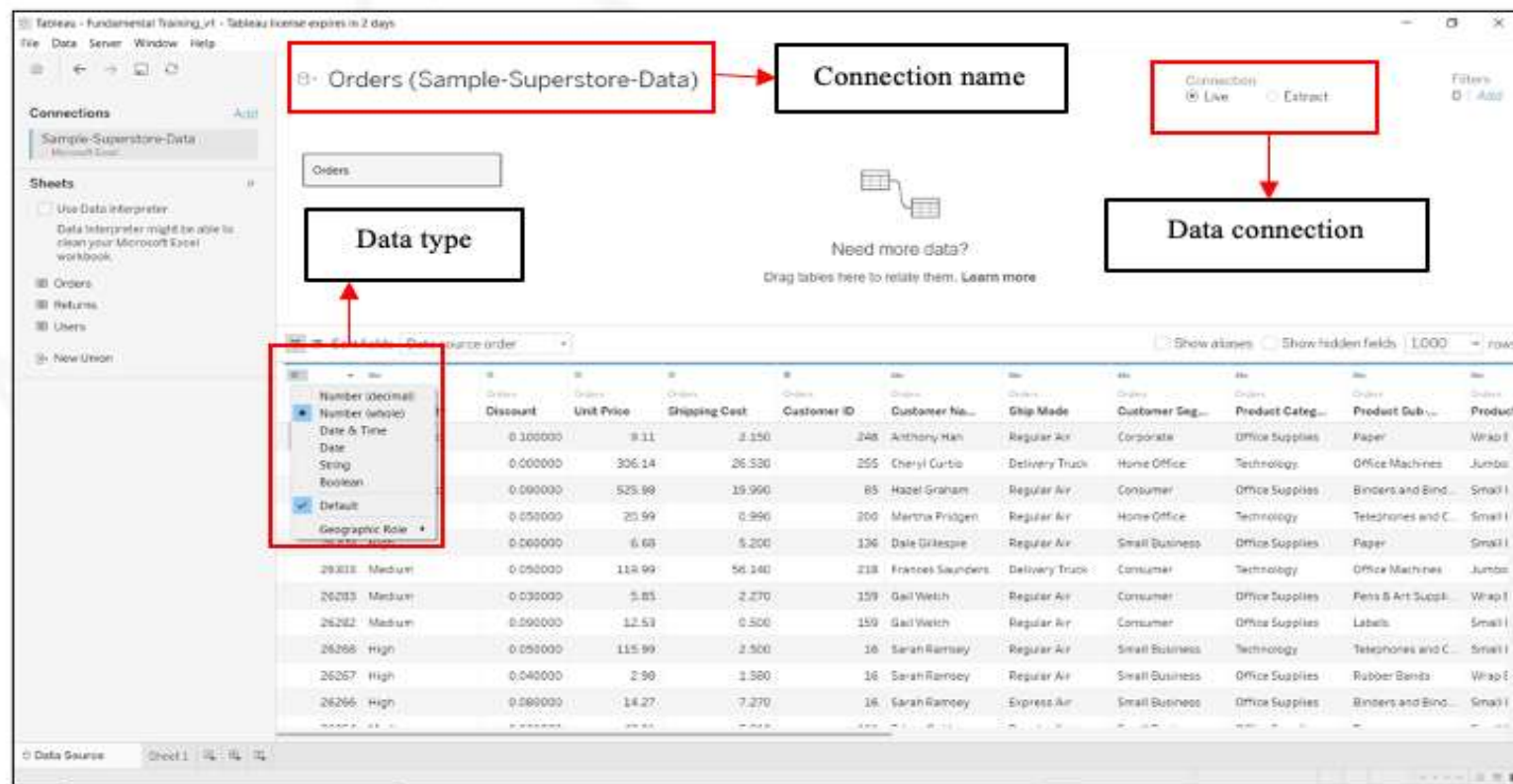


Tableau - Fundamental Training_v1 - Tableau license expires in 2 days

File Data Server Window Help

Connections **Sample-Superstore-Data** Microsoft Excel

Sheets

Use Data Interpreter
Data Interpreter might be able to clean your Microsoft Excel workbook.

Orders
Returns
Users
New Union

Orders (Sample-Superstore-Data) → **Connection name**

Data type

Need more data?
Drag tables here to relate them. Learn more

Connection: ☒ Live ☐ Extract → **Data connection**

Number (Decimal)
Number (Integer)
Date & Time
Date
String
Boolean
Default
Geographic Role

Discount	Unit Price	Shipping Cost	Customer ID	Customer Name	Ship Mode	Customer Segment	Product Category	Product Subcategory	Product Name
0.100000	8.11	3.150	248	Anthony Han	Regular Air	Corporate	Office Supplies	Paper	Wrap I
0.000000	306.14	26.530	255	Cheryl Curbio	Delivery Truck	Home Office	Technology	Office Machines	Junco
0.000000	525.98	19.990	85	Hazel Graham	Regular Air	Consumer	Office Supplies	Binders and Bind	Small I
0.050000	20.99	0.996	200	Martha Pridden	Regular Air	Home Office	Technology	Telephones and C	Small I
0.000000	6.68	5.200	126	Dale Gillespie	Regular Air	Small Business	Office Supplies	Paper	Small I
0.050000	118.99	56.140	218	Frances Saunders	Delivery Truck	Consumer	Technology	Office Machines	Junco
0.030000	5.85	2.270	159	Gail Welch	Regular Air	Consumer	Office Supplies	Pens & Art Suppl	Wrap I
0.000000	12.53	0.500	159	Gail Welch	Regular Air	Consumer	Office Supplies	Labels	Small I
0.050000	115.99	2.500	16	Sarah Ramsey	Regular Air	Small Business	Technology	Telephones and C	Small I
0.040000	2.90	1.380	16	Sarah Ramsey	Regular Air	Small Business	Office Supplies	Rubber Bands	Wrap I
0.080000	14.27	7.370	16	Sarah Ramsey	Express Air	Small Business	Office Supplies	Binders and Bind	Small I

Data Source Sheet 1

Connect To Data (cont'd)

Data Type

Icon	Data type
Abc	Text (string) values
📅	Date values
🕒	Date & Time values
#	Numerical values
T F	Boolean values (relational only)
🌐	Geographic values (used with maps)

Connect To Data (cont'd)

Data Connection (Live Vs Extract)

Live	Extract
Maintains a live connection with the data source.	Extract a copy of the data into Tableau environment.
Any changes in data source will be reflected in Tableau.	No changes unless data extract is being refreshed.
Performance as fast as your database.	Performance depends on your PC's resources.
Secure – User would need to reconnect the data source with credentials.	Not secure because data are not encrypted.
Need to be online if not connecting to local data.	Convenience - can be used offline.

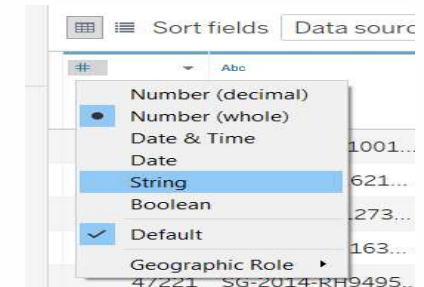
Connect To Data (cont'd)

Data Preparation

▪ Changing Data Type

You can change the data type of a column by clicking the data type icon at the top of the column.

For example, Row ID, although it is a number, but it is not numerical. Change the Row ID to String, by simply click on the icon and choose String.



▪ Changing Name of Data Field

Double clicking on its name and the name of data field can be renamed. For example, double click on the Ship Date, and change the name to Shipping Date. Repeat the same steps for Ship Mode.



Connect To Data (cont'd)

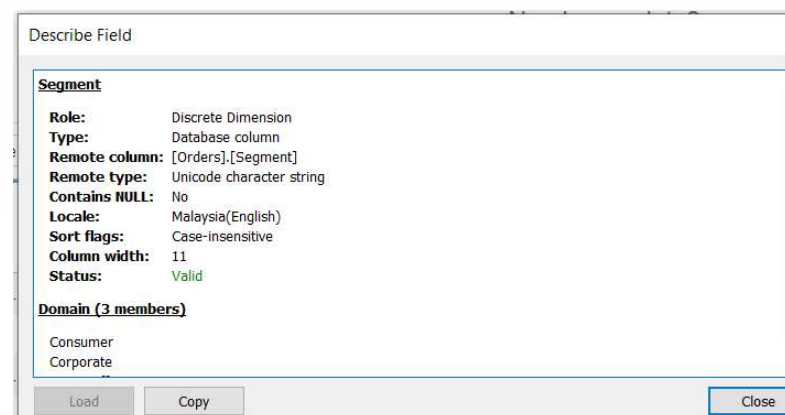
Data Preparation

- **Split**

Right-click on the data field and select Split. For example Order ID .Tableau will detect the hyphen as the separator and split the column into 4 new data fields.

- **EDA (Exploratory Data Analysis)**

Right-click on the data field and select Describe. You can see what is the role has been assigned by Tableau to this data field and the distinct values this column contains. For example select describe for segment field.

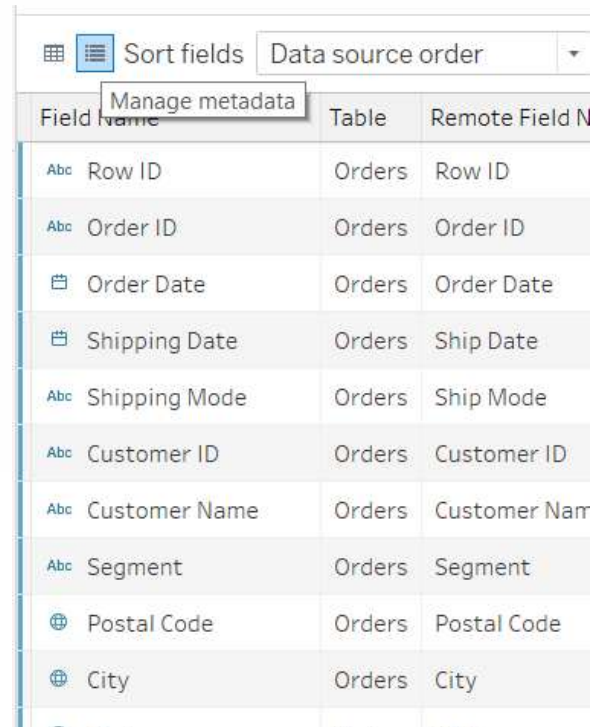


Connect To Data (cont'd)

Data Preparation

▪ Metadata Grid View

This can be a useful view, as the vertical layout can be easier to navigate, especially if you have large number of data fields. Useful when tables have been joined, to know where does each of the data field are actually coming from before joining.



The screenshot shows a software interface for managing metadata. At the top, there are tabs for 'Sort fields' and 'Data source order'. A 'Manage metadata' button is visible. Below is a table with columns: 'Field Name', 'Table', and 'Remote Field Name'. The table lists various fields from an 'Orders' table, including Row ID, Order ID, Order Date, Shipping Date, Shipping Mode, Customer ID, Customer Name, Segment, Postal Code, and City.

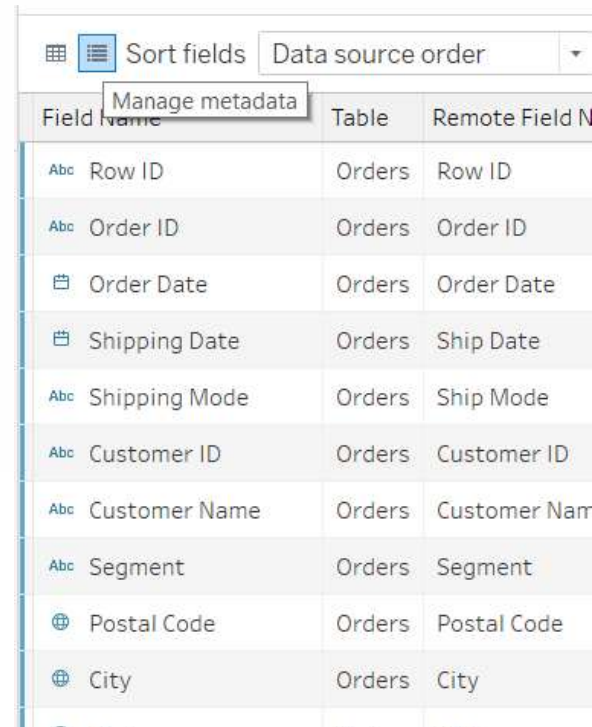
Field Name	Table	Remote Field Name
Row ID	Orders	Row ID
Order ID	Orders	Order ID
Order Date	Orders	Order Date
Shipping Date	Orders	Ship Date
Shipping Mode	Orders	Ship Mode
Customer ID	Orders	Customer ID
Customer Name	Orders	Customer Name
Segment	Orders	Segment
Postal Code	Orders	Postal Code
City	Orders	City

Connect To Data (cont'd)

Data Preparation

▪ Metadata Grid View

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Field Name	Table	Remote Field Name
Row ID	Orders	Row ID
Order ID	Orders	Order ID
Order Date	Orders	Order Date
Shipping Date	Orders	Ship Date
Shipping Mode	Orders	Ship Mode
Customer ID	Orders	Customer ID
Customer Name	Orders	Customer Name
Segment	Orders	Segment
Postal Code	Orders	Postal Code
City	Orders	City

Tableau - Basics & Concepts

Click on **Sheet1** at the bottom left. The new worksheet interface will be prompted..

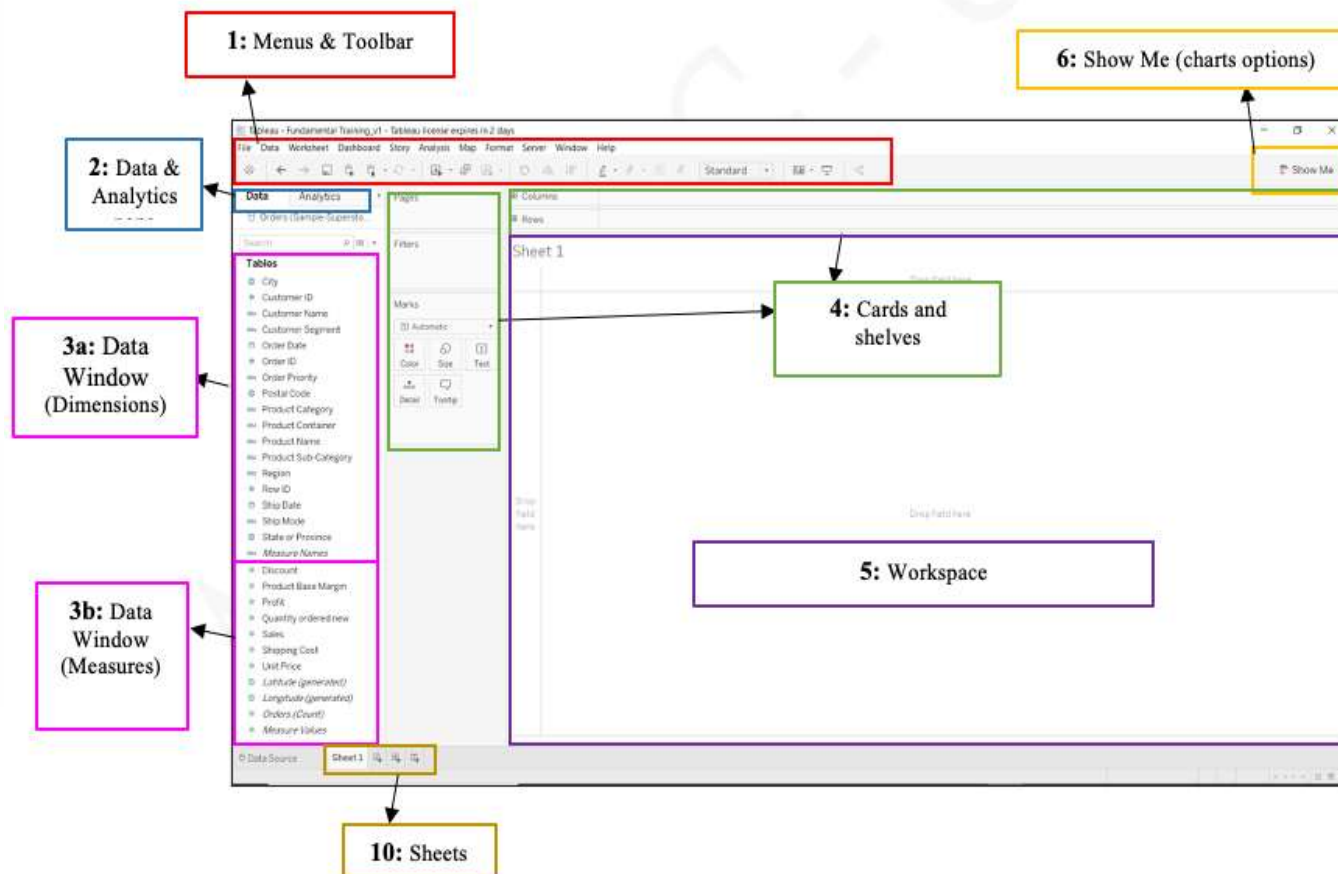


Tableau - Basics & Concepts (cont'd)

Dimension vs Measure

Dimension	Measure
Qualitative (categorical) data	Quantitative (numerical) data
String, Boolean, ID, date and time	Numerical data, latitude and longitude
Usually discrete (blue)	Usually continuous (green)
No aggregation will be applied when the data field	Aggregation will be applied automatically, by default - sum

Tableau - Basics & Concepts (cont'd)

Discrete vs Continuous

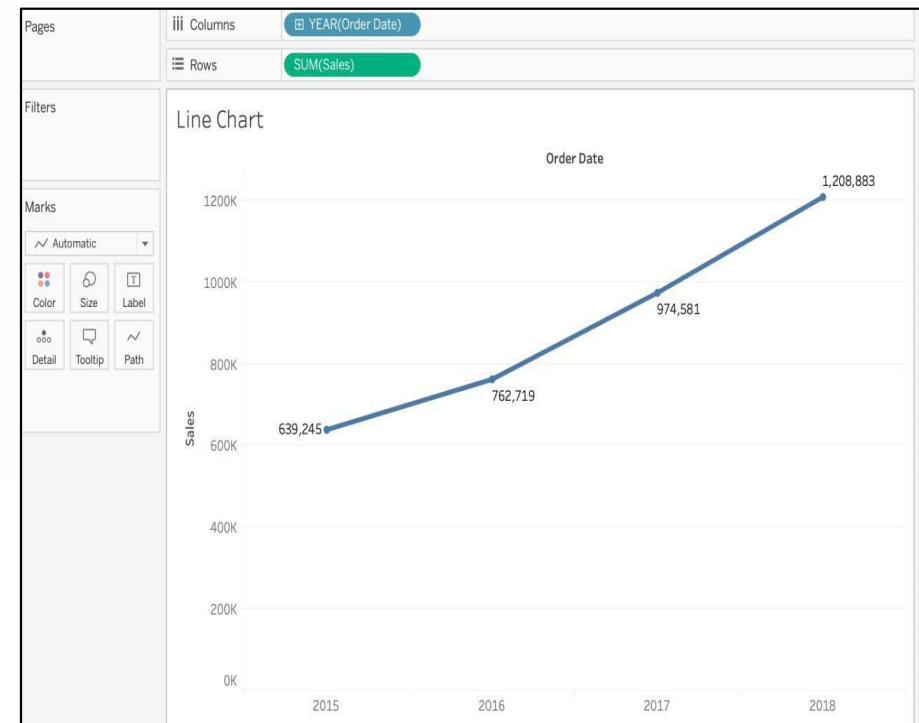
Discrete (blue)	Continuous (green)
Individual, separate values that do not contain values in between.	Can take on any value at any point within the range of values
Is counted	Is measured
Example: No of items in a transaction, no of person in a class	Example: Height, Windspeed
In Tableau, by default, all dimension values are discrete.	In Tableau, by default, all measure values are continuous.

Creating Basic Visualization Charts

Line Chart

Analysis 1: What are the trend of sales over time?

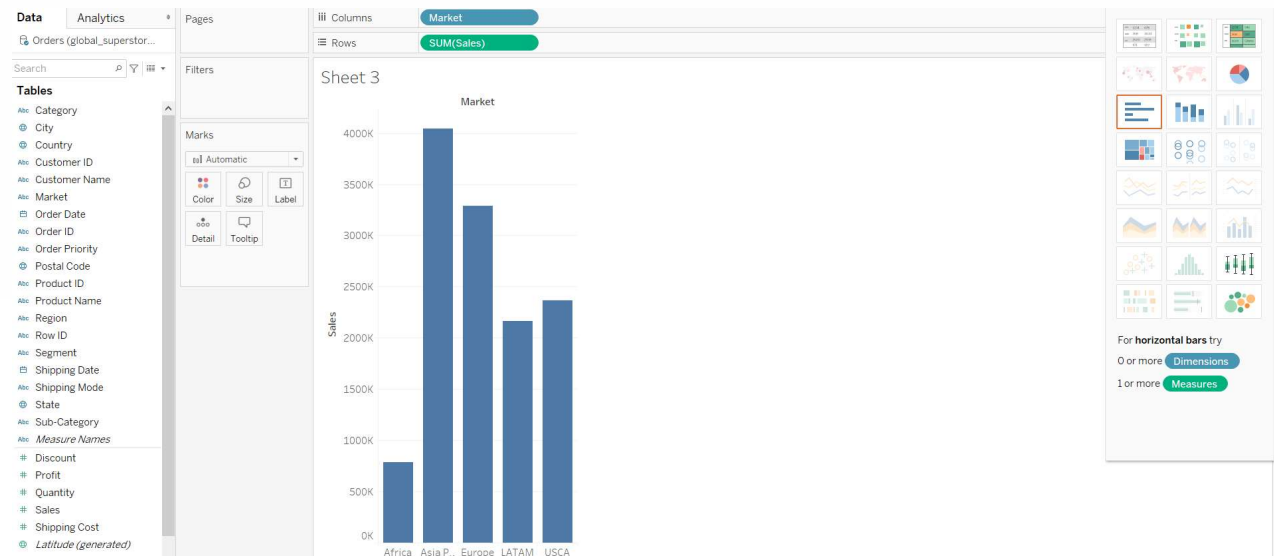
1. In the **Data** pane, select **Sales** and **Order Date**.
2. Go to **Show Me**, click on the recommended chart (highlighted).
3. In the **Marks** card, click **Label** and tick **Show mark labels**.
4. Rename sheet as “Sales over time”.
5. Save your workbook.



Bar Chart

Analysis 2: Which product category has the highest sales?

1. Create a new worksheet.
2. In the **Data** pane, drag **Sales** to **Rows** shelf.
3. Drag **Category** to **Columns** shelf.
4. Go to Sales axis, untick Show Header.
5. In the **Marks** card, click **Label** > tick **Show mark labels**.
6. Optional: To make the visualization fit into your screen, click drop down menu at **Standard** toolbar, select **Entire View**.
7. Rename sheet as "Sales by Category"



Scatter plot

Analysis 3: Is the highest sales has highest profit?

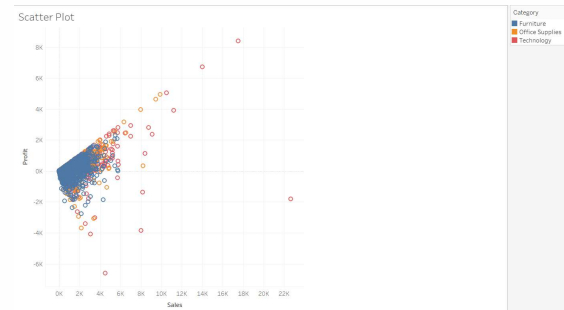
1. Create a new worksheet. Rename sheet as “Sales vs Profit”
2. In the Data pane, highlight **Profit** and **Sales**.
3. Go to **Show Me**, click on **scatter plots**.
4. Go to **Analysis** menu, untick *Aggregate Measures*.



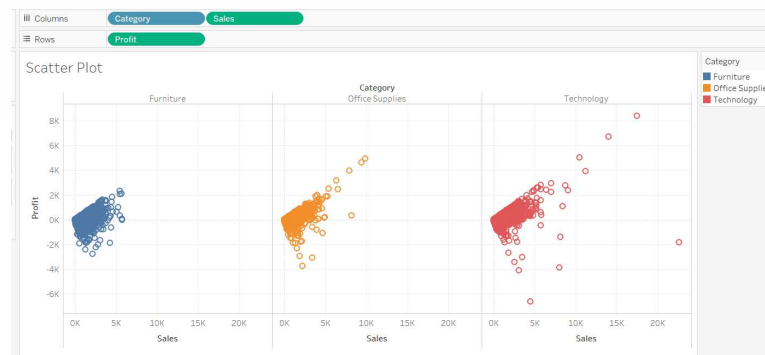
Scatter plot

Analysis 4: What is the Sales and Profit relationship based on category?

1. Duplicate sheet "Sales vs Profit" and rename as Sales vs Profit by Category
2. Drag **Category** data field and drop it in the Color marks card.
The data points are now colored according to their respective Category.



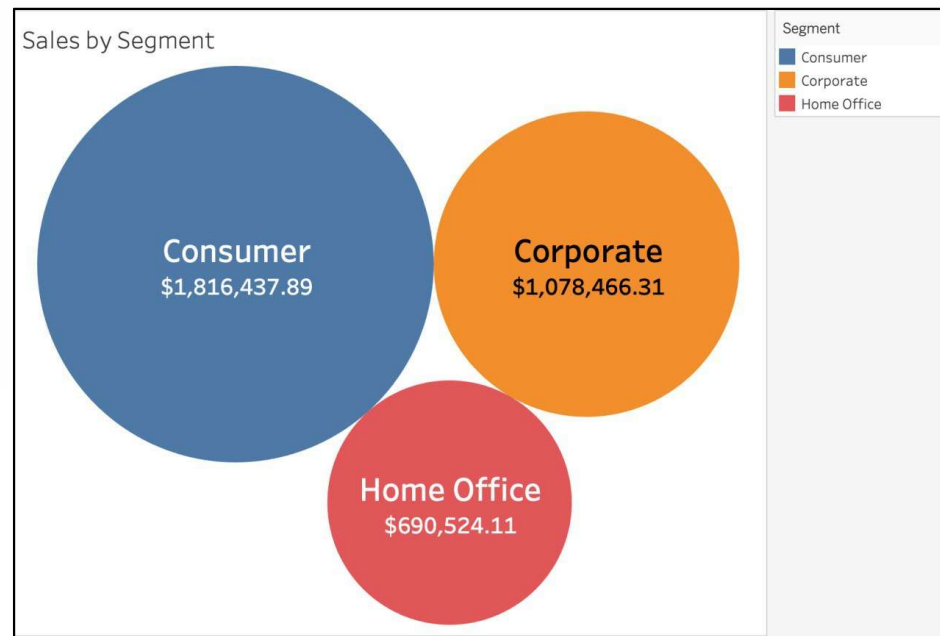
2. The Scatter Plot also can be break down into 3 scatter plot, each for a Category by dragging the Category field and drop it to the Column shelf



Packed-Bubbles

Analysis 5: Which segment has the highest sales?

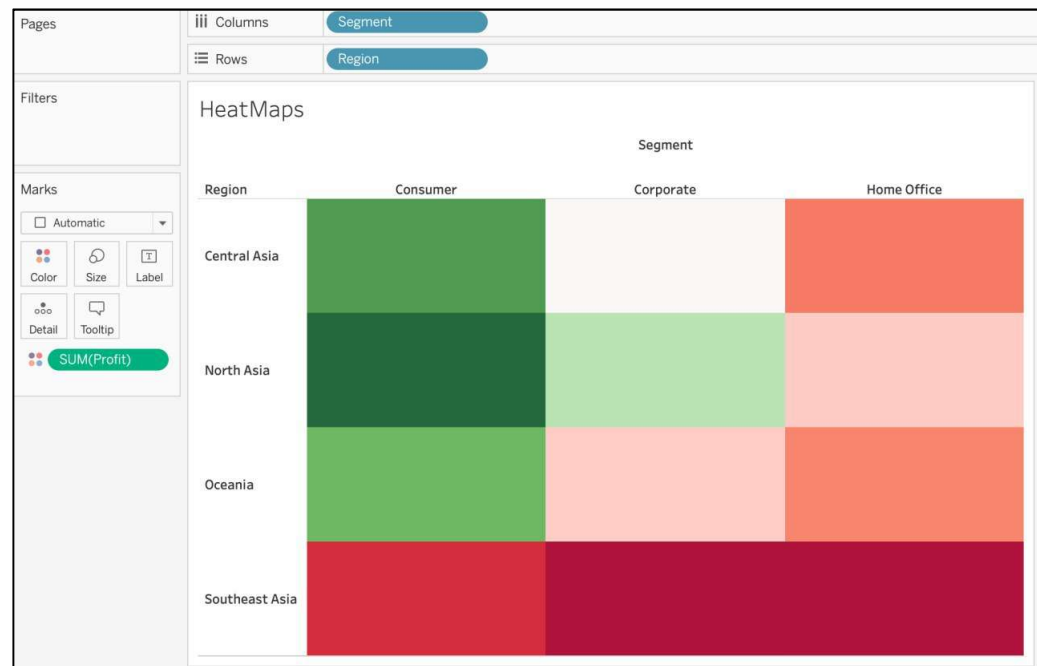
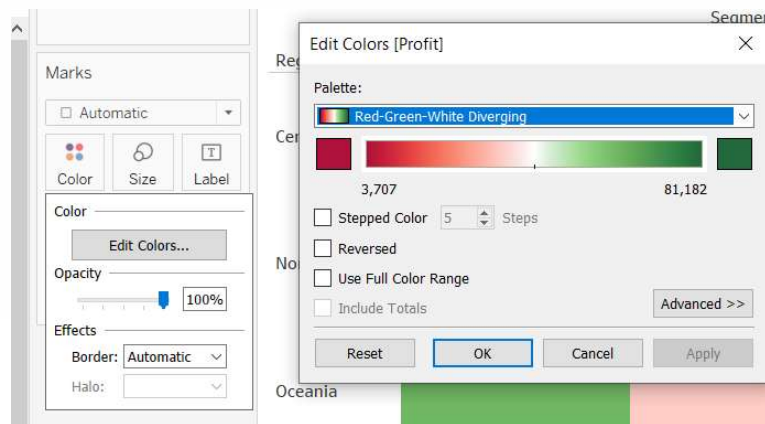
1. Create a new worksheet and rename as “Sales by Segment”.
2. In the **Data** pane, highlight **Sales** and **Segment**.
3. Go to **Show Me**, click on packed bubbles chart.
4. Drag **Sales** to **Label** on **Marks** card.
5. Optional: Click on **Presentation Mode** toolbar or hit function F7 on your keyboard.



Heat Map

Analysis 6: How the profit varies across regions and customer segments?

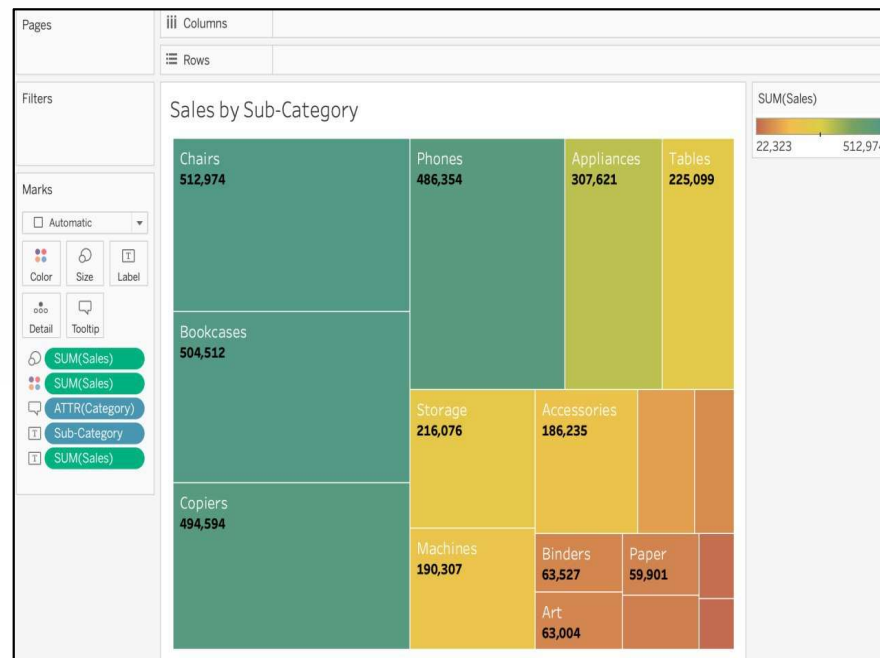
1. Create a new worksheet and rename as "Heat Map".
2. Drag **Segment** to **Columns** shelf.
3. Drag **Region** to **Rows** shelf.
4. Drag **Profit** to **Color** on the **Marks** card.
5. Change **Standard** view to **Entire View**.



Tree Map

Analysis 7: Which product sub-category has the highest sales?

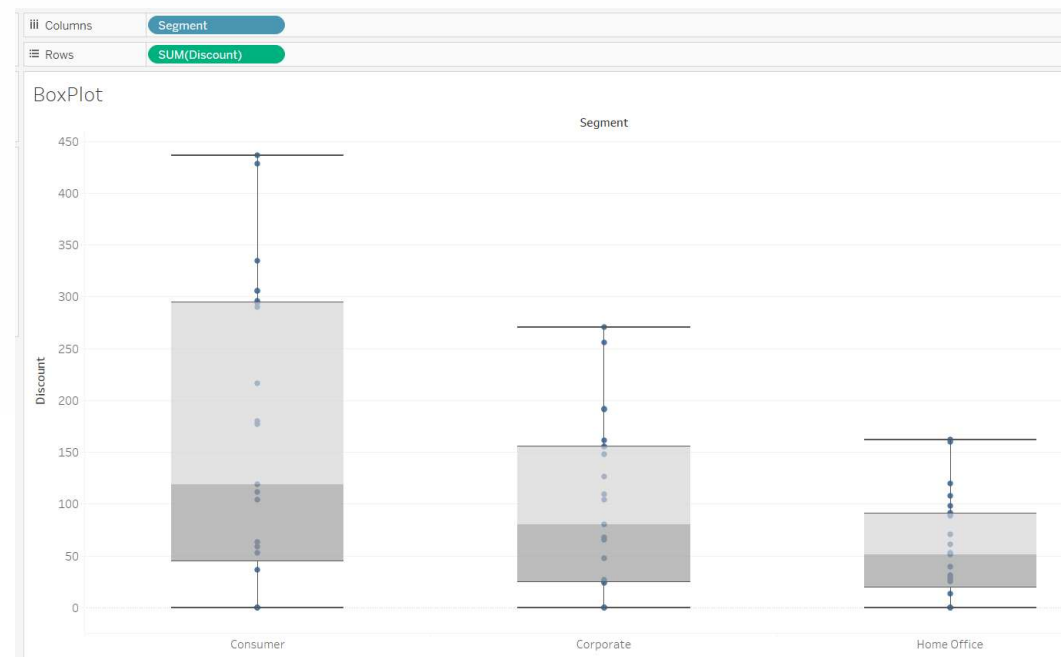
1. Create a new worksheet and rename as Tree Map.
2. In **Data** pane, highlight **Sales** and **Sub-Category**.
3. Go to **Show Me**, click on **treemaps**.
4. Optional: Drag **Category** to **Tooltip** on **Marks** card.
5. Optional: Drag **Sales** to **Label** on **Marks** card.
6. Rename your sheet as "Sales by Sub-Category".



Box Plot

Analysis 8: How the discount varies across regions and customer segments?

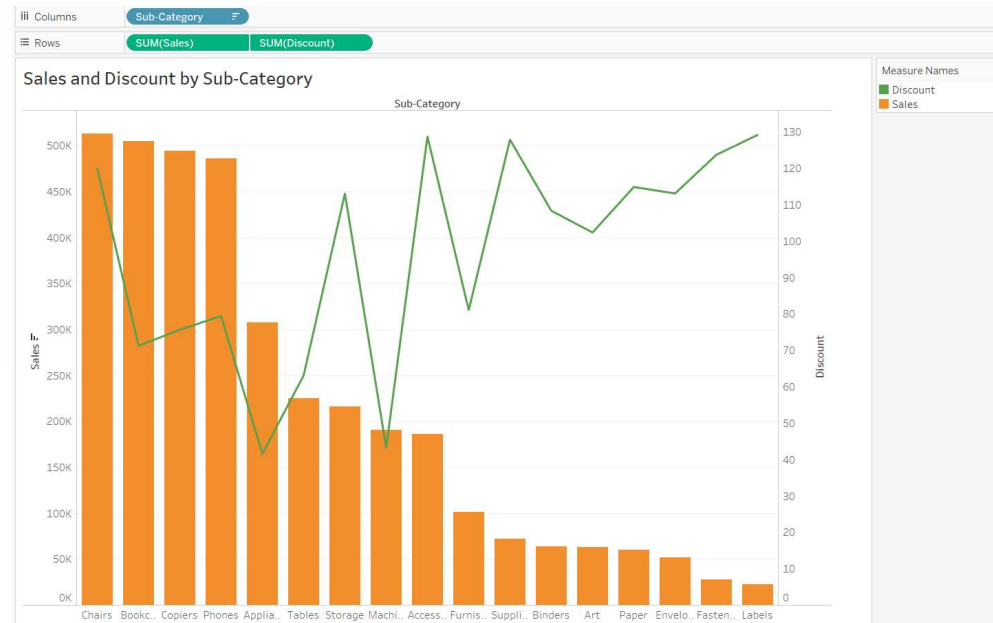
1. Create a new worksheet.
2. Drag **Segment** to **Columns** shelf. Then, drag **Region** to **Columns** shelf and drop it to the right of **Segment**.
3. Drag **Discount** to **Rows** shelf.
4. Click **Show Me** in the toolbar, then select **box-and-whisker plots** chart type.



Combination Charts

Analysis 9: Compare sales and discount by sub-category.

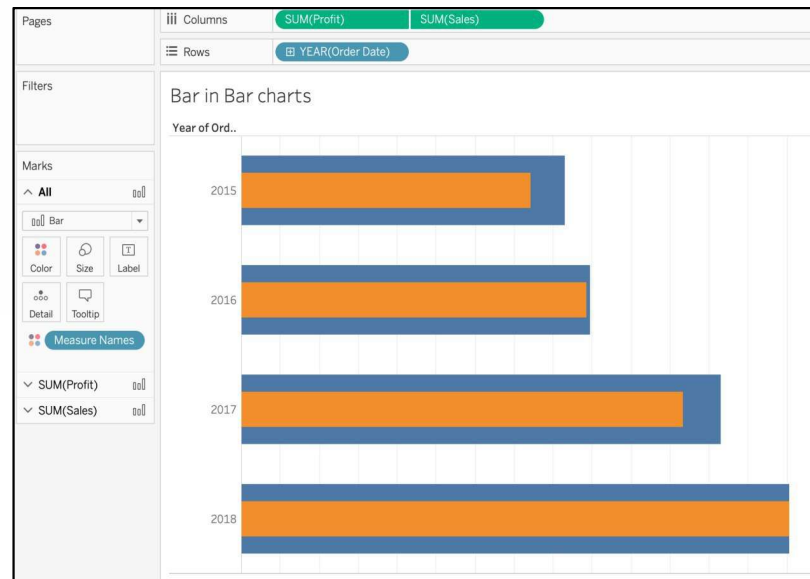
1. Create a new worksheet. Rename it as “Sales and Discount by Sub-Category”
2. On **Data** pane, drag **Sub-Category** to **Column** shelf.
3. Drag **Sales** and **Discount** to **Row** shelf.
4. Click on **SUM(Discount)** pill, select **Dual Axis**.
5. Click on **SUM(Sales)** at **Marks** card and change the chart type to **Bar**.
6. Click on **SUM(Discount)** at **Marks** card and change the chart type to **Line**.
7. Click on the sort icon at the **Sales** axis to sort the data descending by sales.



Overlapped Bar Chart

Analysis 10: Compare sales and profit in one chart.

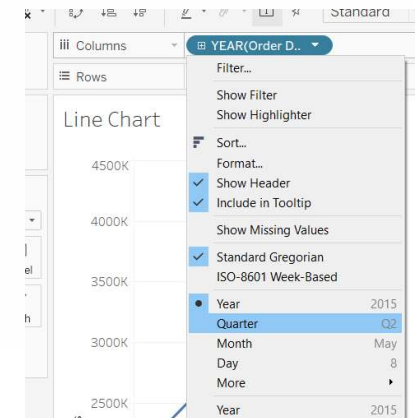
1. Create a new worksheet. Rename as “Bar in Bar charts”.
2. On **Data** pane, drag **Order Date** to **Columns** shelf.
3. Drag **Profit** and **Sales** to the **Rows** shelf.
4. Right-click the second measure on the **Rows** shelf, and select **Dual Axis**
5. On the **Marks** card labeled **All**, set the mark type to **Bar** in the dropdown menu.
6. On the **Marks** card labeled **SUM(Sales)**, click **Size** and then adjust the slider to change the width.
7. Repeat step 6 on the Marks card labelled **SUM(Profit)**.



Organizing Data

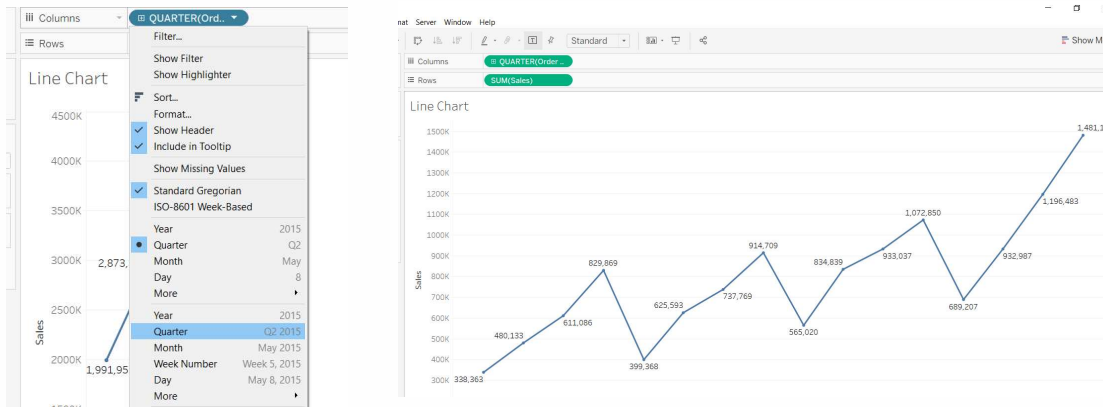
Hierarchies and Drill-Down-Date

- Duplicate the “Sales Over time” sheet and name it as “Continuous date”.
- Tableau displays Order Date at the year level by default.
- In Tableau, for data fields with date values, Tableau will automatically create hierarchy on dates from year to month to quarter and then finally day.
- We can click the plus sign and continue to drill down through the hierarchy process.
- We can click on the minus sign on the year pill to drill back up.
- If you want to change the year level directly to quarter, from the Order Date pill, click on the dropdown button and choose Quarter – Q2
- Blue colour for the Order Date pill on column shelf indicating discrete data type.
- Q1 point, a Q2 point, a Q3 point and Q4 point for every year in the data source all rolled up into single marks.
- As in the Q1 Sales we are looking in this chart here, is actually the sum of Q1 Sales for 2012, 2013,14 and the year 2015 added up.



Data part versus date value

- At column shelf drop-down menu on the month of Order Date, select the second Quarter Q2 2015 entry in the menu.



- The chart changes and the quarter of Order Date turns green. Dates can be treated as either discrete or continuous.
- Thus, blue does not indicate Dimensions and green does not indicate Measures. Actually, blue indicates discrete fields while green indicates continuous fields.
- Now see a very different chart with 16 marks starting in Q1 2012, ending in Q4 2015 and the date range becomes an axis

Hierarchies

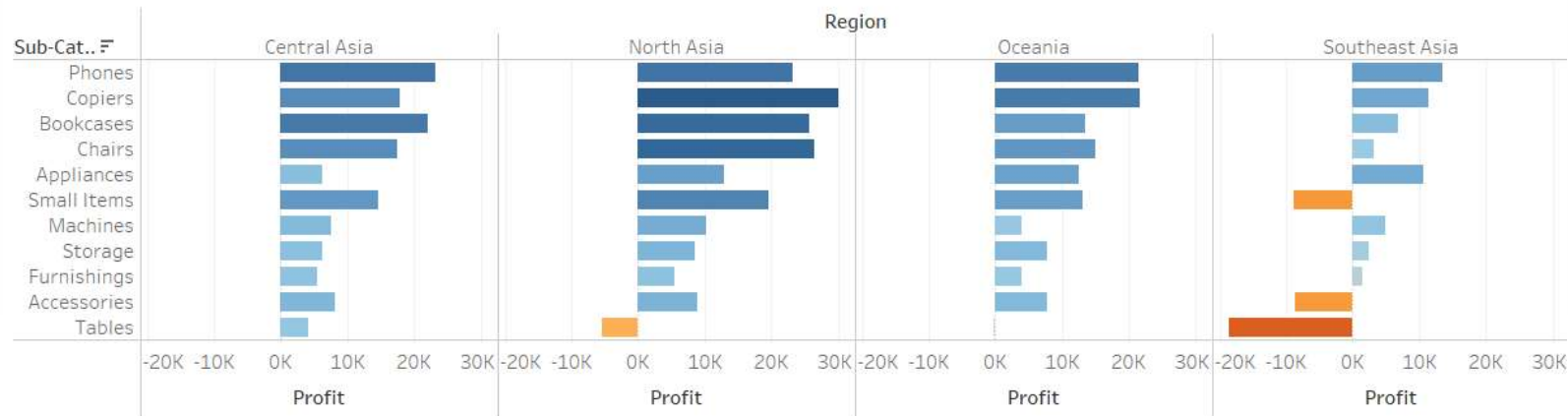
1. In the **Data** pane, drag **Sub-Category** and drop it directly on top of **Category** field.
2. When prompted, enter a name for the hierarchy and click **OK**.
3. Drag additional fields into the hierarchy as needed. You can also re-order fields in the hierarchy by dragging them to a new position.

Sorting

1. Go to **Combination Chart** sheet.
2. Hover to the Sales axis and click the Sort icon.
3. Another way to in ascending order is from the Ascending Sort button from the Toolbar. Click on another button beside it if you want to sort it in descending order.
4. Right-click on **Sub-Category** pill. Select **Sort**. The Sort dialog box as below will be prompted. Do configuration as follows: **Sort By** > select **Field**; **Sort Order** > **Descending**

Grouping

1. On **Data** pane, drag **Sub-Category** to **Row** shelf.
2. Drag **Profit** and **Region** to **Column** shelf.
3. In **Data** pane, drag **Profit** to **Color** on the **Marks** card.
4. Hover to **Office Supplies**. Click on **Binders**.
5. Hit **shift** button on your keyboard, then click the last item, i.e., **Fasteners**
6. Right-click on highlighted items, select **Group**.
7. To rename the group, right-click on that grouped items, click Edit Alias.
8. In the Edit Alias dialog box, type new name for the group. For example, "Small Items".



Filtering

We use filter to select a subset of the data in the view to focus on that data or to exclude data from the view.

Filter shelf

1) Drag the **Region** field to the **filter** shelf, market. Tick the Central Asia and

Interactive Filter

1) Right-click on the **Region** field in the data pane, click Show Filter. Notice there will be a filter box appears on the right of the chart.

2) User can interact with the chart easily by clicking on the filter to choose which market to include on the Bar Chart.

3) If you want to allow the user to only click a single market at once,

- Go to the filter box on the right,

- Click on the drop-down menu.

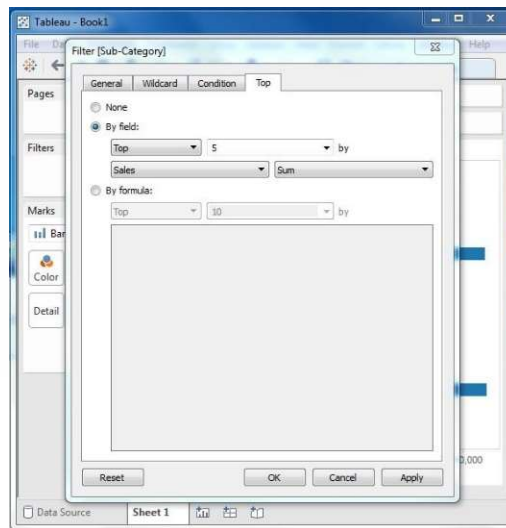
- Click on the Single value list.

Note that the filter now has turn into a radio button

Top Filter

Analysis: View top 5 products

- 1) Create a new worksheet. Rename it as “Top 10 Product”
- 2) Drag **Sub-Category** to the Rows shelf and the **Sales** to the Columns shelf. Choose the horizontal bar as the chart type.
- 3) Right-click on the field Sub-Category and select filter.
- 4) Go to the tab named Top. Here, choose the second radio option by field. From the drop-down, choose the option Top 5 by Sum of Sales

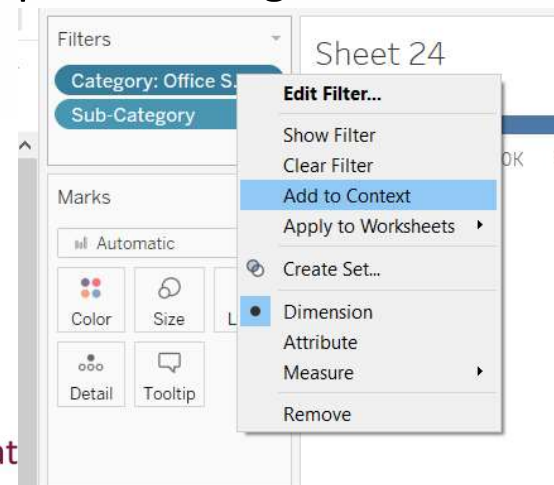


Context Filters

Analysis: View top 5 Sub-Category of products for the category called Office Supplies.

The normal filters in Tableau are independent of each other. It means each of the filter reads all the rows from the source data and creates its own result. However, there may be scenarios where you might want the second filter to process only the records returned by the first filter. In such a case, the second filter is known as dependent filters because they process only the data that passes through the context filter.

- 1) Drag the dimension Category to the filter shelf.
- 2) Choose Office Supplist from the list.
- 3) Right-click the Category: Office Supplies filter and select the option Add to Context. This produces the final result, which shows the subcategory of products from the category Office Supplies which are among the top 5 subcategories across all the products.

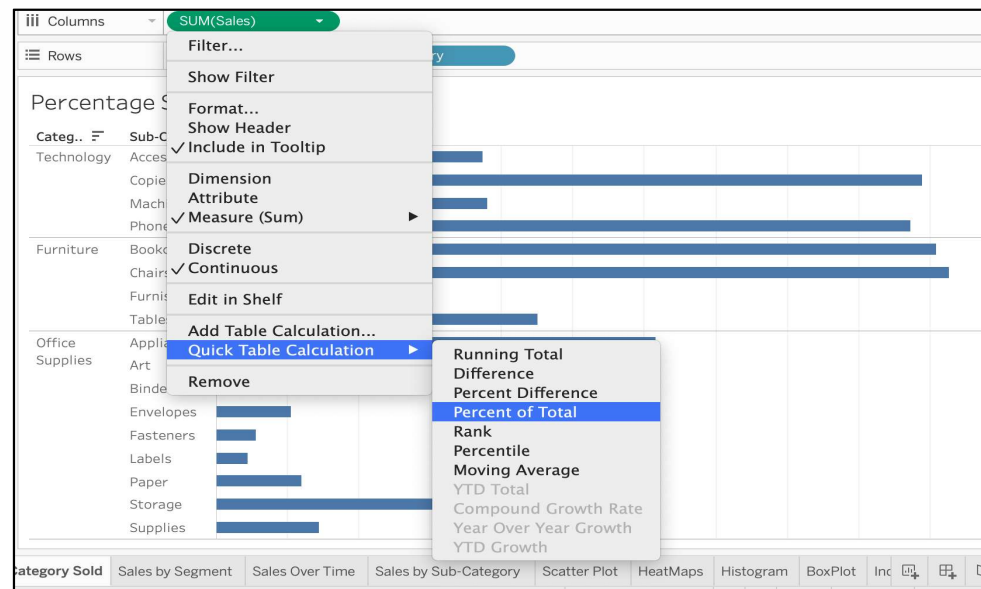


Analysis 11: How many percent of customers that has sum of sales over \$5000?

1. Create a new worksheet. Rename it as “Set”.
2. In the **Data** pane, right-click on **Customer Name** > select **Create** > select **Set**.
3. In the Create Set dialog box,
 - a. Type a name for the set. Let's put “Customers A”.
 - b. On **General** tab, click on **Use all** button. This is to make sure the condition always applies to all values even when new customers are added.
 - c. Hit the **Condition** tab. Click **By field** button. Then, define a condition that only includes customers when sum of sales is greater than 5000.
 - d. Click **OK**.
4. In the **Data** pane, drag the **Customers A** to **Row** shelf. Then, drag **Sales** to **Columns** shelf.
5. On the **SUM(Sales)** pill, right-click and select **Quick Table Calculation** > **Percent of Total**.
6. Click **Label** on the **Marks** card, tick **Show marks label**.

Quick Table Calculation

1. Go to **Bar Chart** sheet. Duplicate this sheet. Rename as “Percentage of Sub-Category Sold”
2. In the **Data** pane, drag **Sub-Category** to **Columns** shelf and drop it after **Category** pill.
3. Swap the columns to row by clicking Swap icon  at the toolbar.
4. Right-click the **SUM(Sales)** pill, select **Quick Table Calculation** > select **Percent of Total**. The marks label will be changed from aggregation sum to percentage.

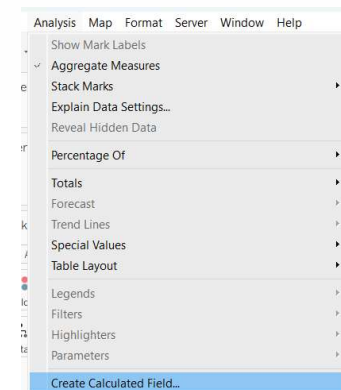


Calculated Field

If the data source that we use does not contain a field (or column) that we need for analysis, then we can use create calculated field function to create a create new dimensions such as segments or new measures such as ratios.

Analysis 13: How many customers (as identified by Customer ID) made 4 or 5 separate orders?

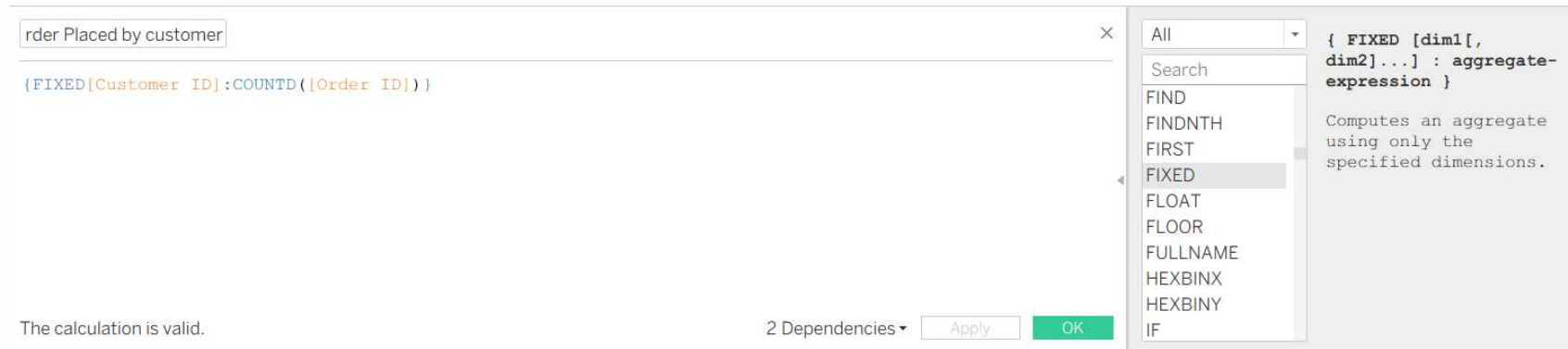
1. Since data source does not contain a field (or column) for this analysis (i.e., no. of orders by customer ID), therefore we need to create the new measure using **Create Calculated Field** function that "fixes" the calculation at the customer level using **FIXED** level of detail (LOD) expressions and **counts** their distinct orders.
2. Create a new worksheet. Rename it as "Histogram".
3. Go to Analysis menu, select **Create Calculated Field**.



Calculated Field

4. In the Calculation Editor that opens:

- Enter a name for the calculated field as 'Order placed by customer'
- Enter a formula:



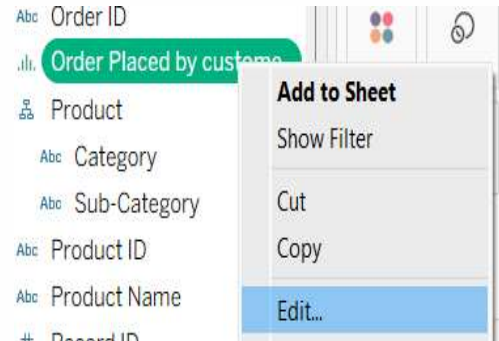
Tip: To see a list of available functions, click the triangle icon on the right-side of the Calculation Editor.

- The "D" in COUNTD() stands for Distinct. This function counts the number of distinct orders(Order Id) for each customer(Customer Id).
- The curly braces {} indicate a fixed level of detail expression, which disregards other dimensions in the view.

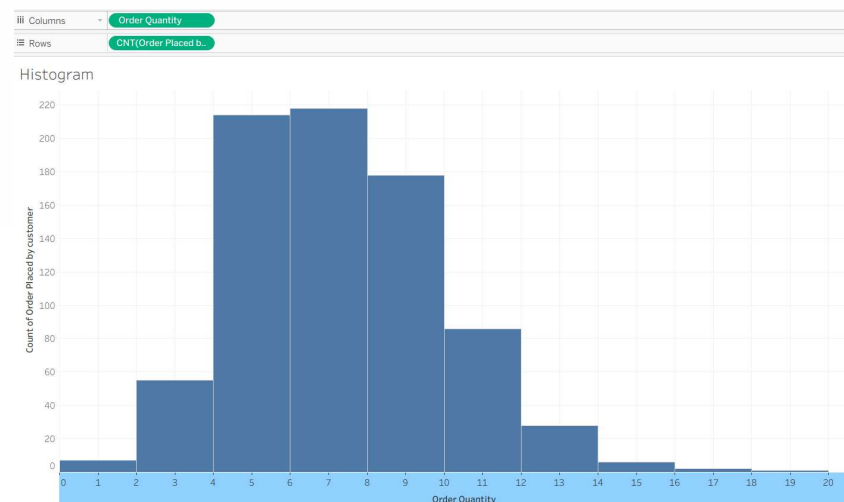
c. When finished, click **OK**. The new calculated field is added to Data pane as a measure because it returns a number. An equal sign (=) appears next to the data type icon.

Calculated Field

5. Drag **Order placed by customer** to **Rows**.
6. Go to **Show Me**, click on **histogram**.
7. To change the bin details, right-click on **Order placed by customer (bin)** in the **Data** pane and select **Edit**



8. In the Edit Bins dialog box:
 - a. Rename the Bins field as Order Quantity
 - b. Change the desired bin size to 2.
 - c. Click **OK**.



Calculated Field

Analysis 14: How the discount ratio varies across regions and product sub-category

1. Create a new worksheet. Rename it as “Discount ratio”.
2. Create a new calculated field measure discount ratio using **Create Calculated Field** function
3. In the Calculation Editor that opens, do the following:
 - a. Enter a name for the calculated field as **Discount Ratio**.
 - b. Enter a formula.

Discount Ratio

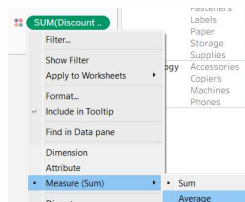
```
IF [Sales] != 0 THEN [Discount]/[Sales] ELSE 0 END
```

This formula checks if sales is not equal to zero. If true, it returns the discount ratio (Discount/Sales); if false, it returns zero.

- c. click **OK**

Calculated Field

4. From the **Data** pane, drag **Region** to the **Columns** shelf.
5. From the **Data** pane, drag **Category** to the **Rows** shelf.
6. On the **Rows** shelf, click the plus icon (+) on the **Category** field to drill-down to Subcategory
7. From the **Data** pane, drag **Discount Ratio** to **Color** on the Marks card
8. On the Rows shelf, right-click **SUM(Discount Ratio)** and select **Measure (Sum) > Average**.



The view updates to show the average of discount ratio.



Session 3:

Application of Data Analytics with Tableau (Part 2)



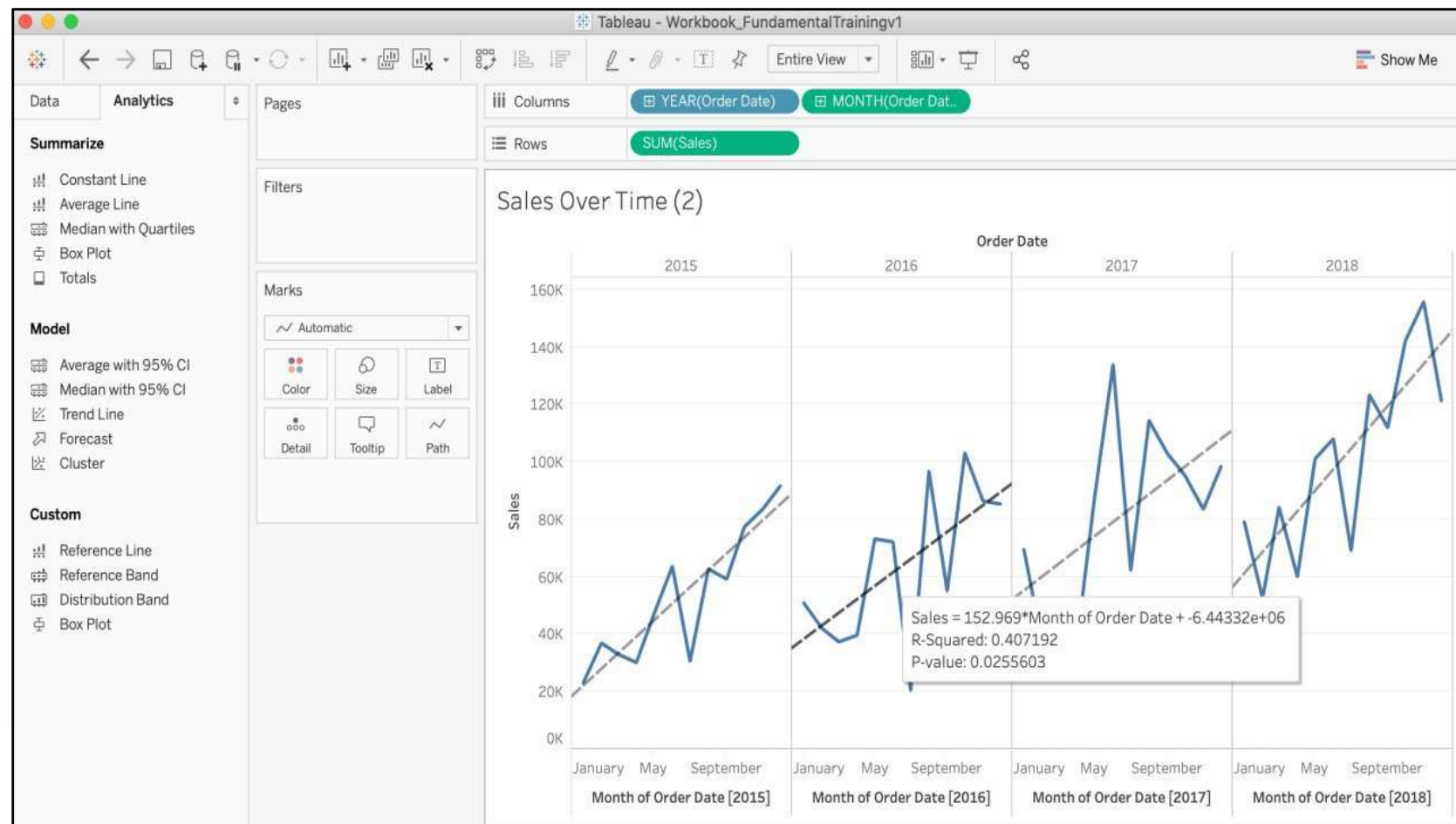
Analytics Tab

Trend Line

Analysis 13: How the relationship of sales over time?

1. Go to “**Sales over Time**” sheet. Duplicate this sheet and rename as “Trend Sales over Time ”
2. Click ‘+’ on the **Year(Order Date)** pill to drill-down till **Month(Order Date)**. Bring out the **Quarter (Order Date)**.
3. In the **Analytics** pane, drag the **Trend Line** to the view and a few possible trend line types to choose will be turned on (Linear, Logarithmic, Exponential, Polynomial, or Power model types).
4. Drop it on the **Linear** type to create a linear trend line in the line chart.
5. Hover over the trend line to display the statistics information including the equation, R-squared and P-values.

Trend Line (cont'd)



Forecasting

Analysis 14: What is the sales forecast value for the next 6 months?

1. Go to **Sales Over Time** sheet. Duplicate this sheet and rename as “Forecast”.
2. Right-click on the trend line and untick **Show Trend Lines**.
3. In the **Analytics** pane, drag Forecast to the canvas. You will see the line chart will be expanded with the forecast values shown by a light blue shaded color. Tableau will automatically forecast up to next 13 months.
4. If you want to change the default configuration for the forecast, right-click on the forecast line, select **Forecast** > click on **Forecast Options**.
5. In the **Forecast Options** dialog box, do the configuration as you wish.

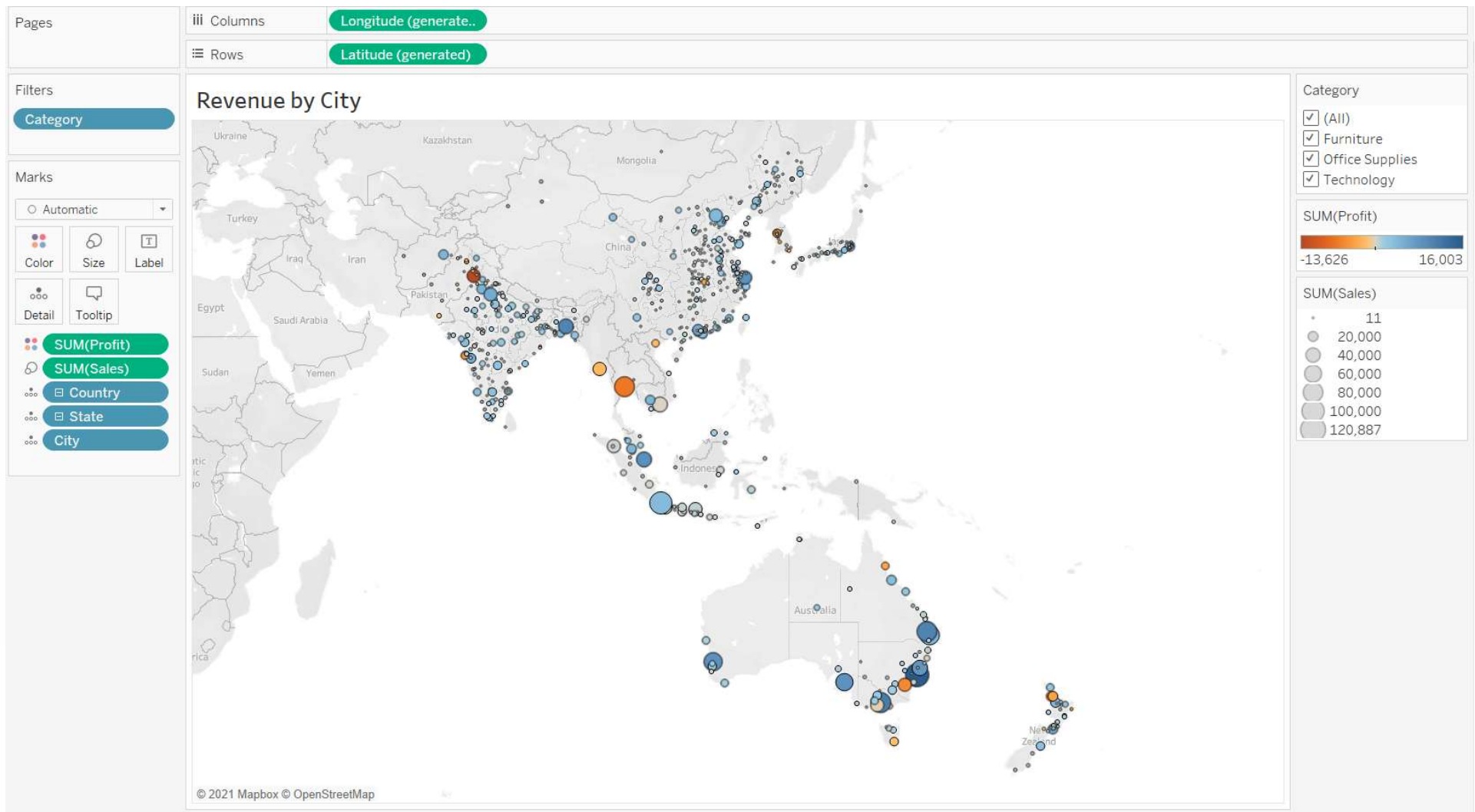
Mapping the Data

Map Data Geographically

Analysis 15: Where is data coming in from? Which state has the highest lost?

1. Create a new worksheet.
2. In the **Data** pane, double click on **State**. Note that, the generated **Latitude** and **Longitude** were automatically plotted on the **Rows** and **Columns** shelf.
3. Drag **Profit** to **Color** on **Marks** card. This should create a filled map colored on a gradient of 'SUM(Profit)'. Note that the default aggregation is SUM.
4. Hover over the darkest orange state to see which state has the highest lost.
5. Then, drag **Sales** to **Size** on the **Marks** card.
6. Resize the shape of the circle by click **Size** and then adjust the slider to change the width.
7. Optional: To put border on the circle, click on **Color** at the **Marks** card, select **Border** > choose color.
8. Optional: You can extend you analysis by adding another measure to **Filter** card. For example, Category. On the **Data** pane, right-click on **Category** field and select **Show Filter**.
9. Optional: You can expand **State** to **City** by clicking the '+' symbol on **State** pill on **Marks** card.
10. Rename this sheet as "Revenue by City".

Map Data Geographically (cont'd)



Dashboard & Stories

Create Dashboard

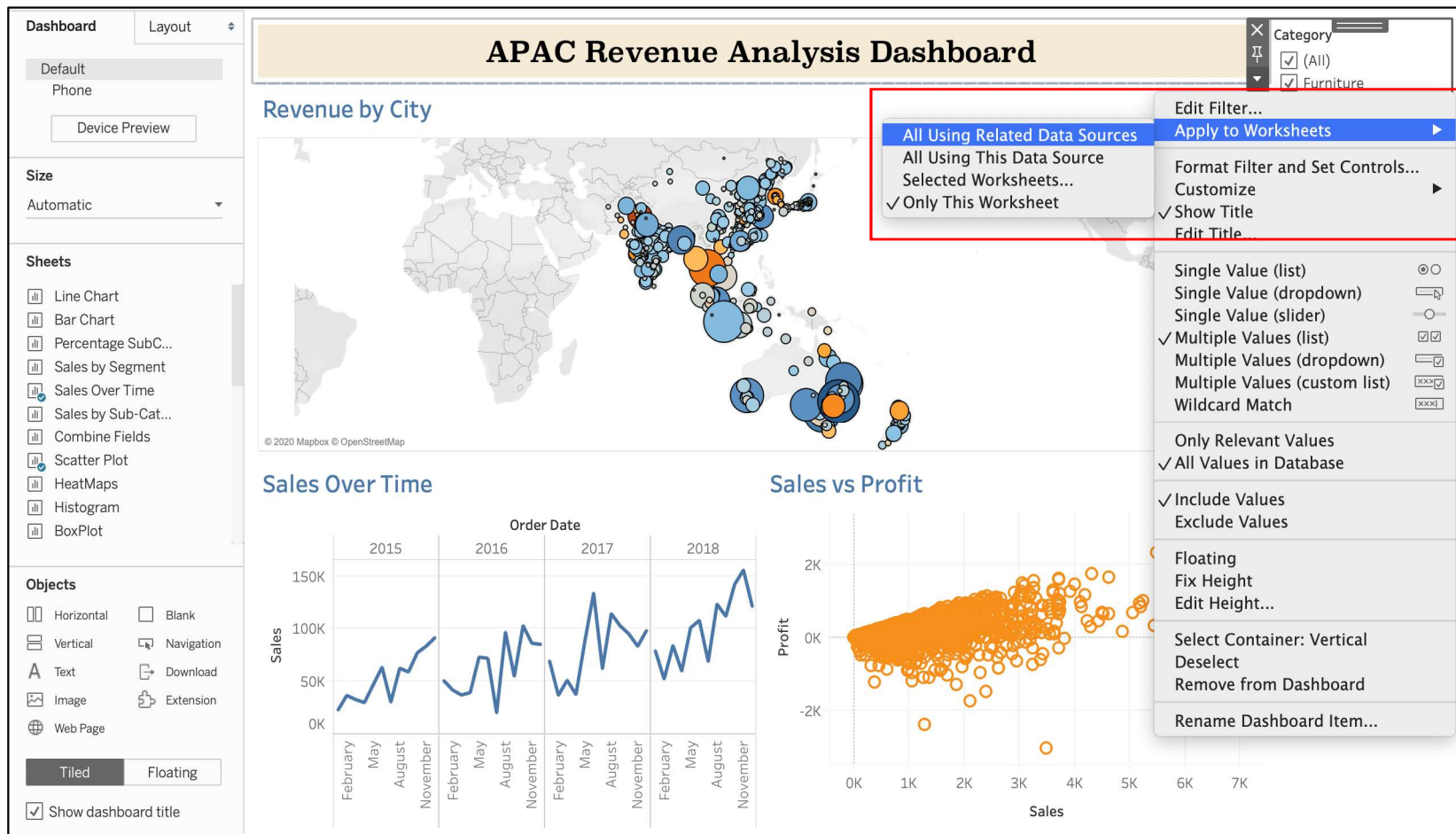
1. Click the **New Dashboard** icon located at the bottom of the workbook. Rename as “Sales Dashboard”.
2. In the **Dashboard** pane, tick on **Show dashboard title**. (Optional)
Change size to **Automatic**.



Create Dashboard (cont'd)

3. From the **Sheets** list at left, drag **Revenue by City** to your dashboard at right.
4. Next, drag **Sales Over Time** sheet to the bottom. Note: As you drag the sheet around the dashboard, a grey shaded area indicates the various places you can drop it.
5. Drag **Scatter Plots** to the bottom right. Rename the title as “Sales vs Profit”.
6. To add interactivity to dashboard to enhance users' data insights, we can use filter as one the options. To configure this, click the caret on **Category** card to bring up the menu. Then, select **Apply to Worksheets** > select **All Using Related Data Sources**

Create Dashboard (cont'd)

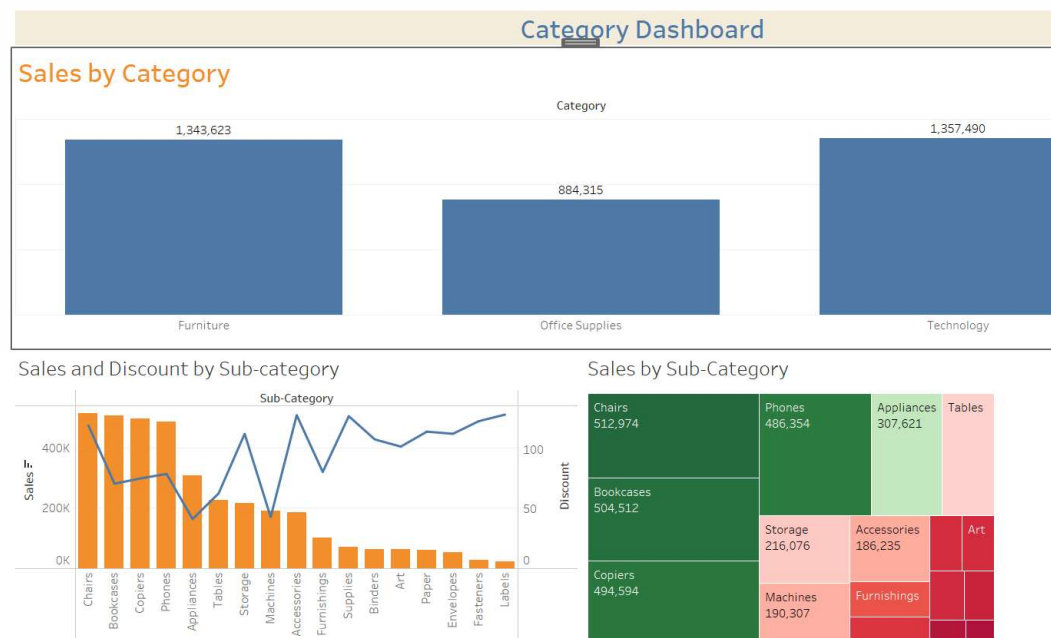


Create Dashboard (cont'd)

Before we proceed to the next sub-topic, kindly create another dashboard that contains the following worksheet:

- a) Sales by Category (use Bar Chart worksheet).
- b) Sales and Discount by Sub-Category
- c) Sales by Sub-Category

Set Sales by Category as a filter. Rename dashboard as “Category Dashboard”.



Create Stories

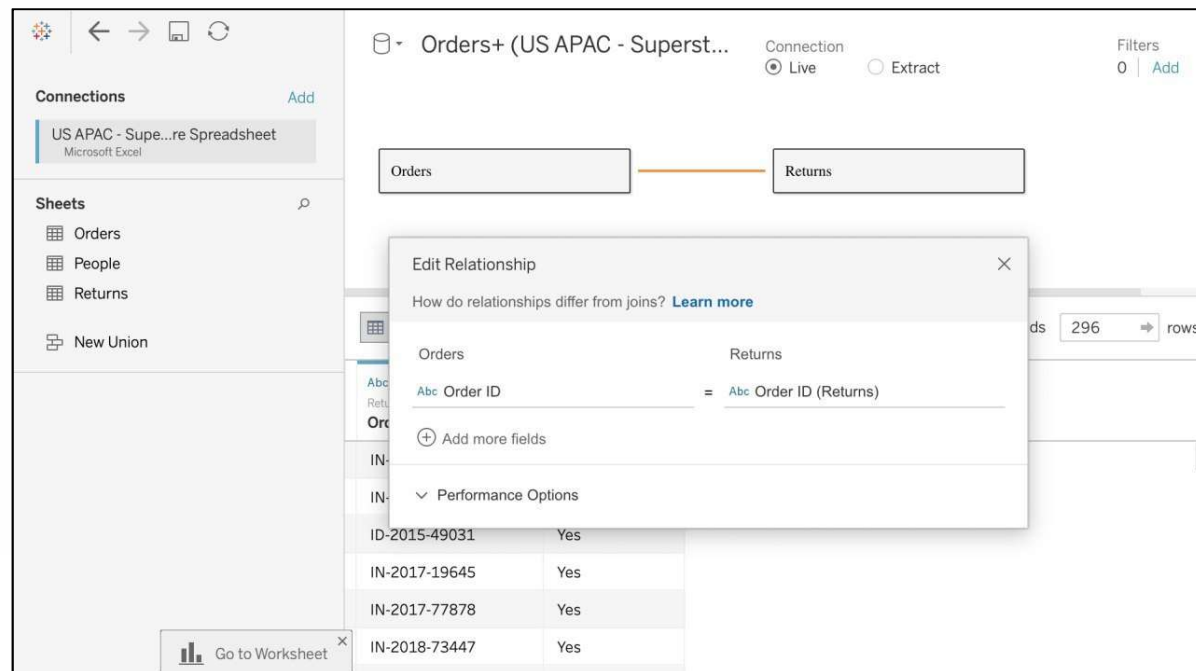
1. Click the **New Story** icon located at the bottom of the workbook.
2. In the **Story** workspace, right-click on the Story 1 title, select **Edit title**.
3. In the Edit Title dialog box, highlight the whole <Sheet Name> and change to “APAC Superstore Revenue Analysis”.
4. In the **Story** pane, drag **Sales Dashboard** to the story at right.
5. Click on the **Navigator box** to add a caption on the point. For this example, write “This is the overall sales analysis in APAC region”. You can re-format the font type, size, etc. by clicking the **Story** menu, select **Format**.
6. Add new story point by clicking **Blank** at Story pane.
7. Drag **Category Dashboard** to the canvas. Add a caption to tell the story.



Data Blending

Single Connection

1. On your workbook, go to **Data Source** window.
2. In the **Sheets** pane, drag another table (i.e., **Returns sheet**) to the canvas. When you see the “noodle” between the two tables, drop that table.



Multiple Connections

1. On your workbook, go to **Data Source** window.
2. In the **Connections** pane, click **Add**.
3. Connect to another data source (i.e., APAC Superstore Spreadsheet (Returns).csv).

Tableau - Combine Data by relationship (multiple connections)

Orders+ (Multiple Connections)

Connection

Live

Extract

Filters

0

Add

Connections

Add

US APAC - Superstore Spreadsheet

Microsoft Excel

US APAC - Superstore Spreadsheet (Returns)

Text file

Sheets

Orders

People

Returns

New Union

Sort fields

Data source order

Show aliases

Show hidden fields

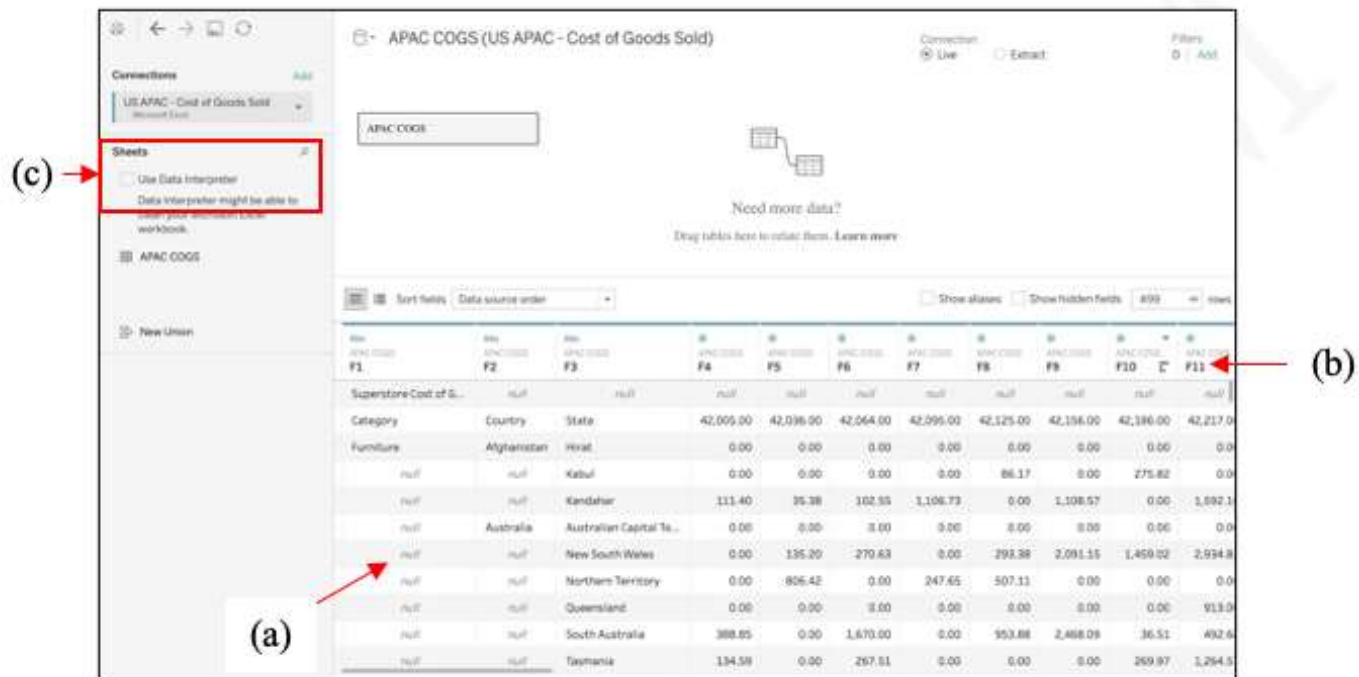
1,000

rows

# Orders Record ID	ABC Orders Order ID	 Orders Order Date	 Orders Ship Date	ABC Orders Ship Mode	ABC Orders Customer ID	ABC Orders Customer Name	ABC Orders Segment	 Orders City
1	IN-2017-27828	26/08/2017	30/08/2017	Standard Class	VT-21700	Valerie Takahito	Home Office	Luoyang
2	IN-2017-27828	26/08/2017	30/08/2017	Standard Class	VT-21700	Valerie Takahito	Home Office	Luoyang
3	IN-2017-27828	26/08/2017	30/08/2017	Standard Class	VT-21700	Valerie Takahito	Home Office	Luoyang
4	IN-2017-27828	26/08/2017	30/08/2017	Standard Class	VT-21700	Valerie Takahito	Home Office	Luoyang
5	IN-2018-63178	24/06/2018	30/06/2018	Standard Class	SO-20335	Sean O'Donnell	Consumer	Marik
6	IN-2015-29067	30/05/2015	03/06/2015	Second Class	ST-20530	Shui Tom	Consumer	New U
7	IN-2016-15361	15/08/2016	21/08/2016	Standard Class	CS-12130	Chad Sievert	Consumer	Tokyo
8	IN-2015-79397	03/01/2015	03/01/2015	Same Day	KN-16450	Kean Nguyen	Corporate	Sydney
9	IN-2015-79397	03/01/2015	03/01/2015	Same Day	KN-16450	Kean Nguyen	Corporate	Sydney

Blend

1. Create a new worksheet.
2. Drag **Profit** to Rows shelf. Then, drag **Category** to **Color** on **Marks** card.
3. To add another data source, click on  icon in the Toolbar.
4. Choose **More...**, and navigate to the MS Excel file: *US APAC - Cost of Goods Sold*, and click **Open**. The preview of data will be prompted as follows.



APAC COGS (US APAC - Cost of Goods Sold)

Connections: US APAC - Cost of Goods Sold (Microsoft Excel)

Sheets: ☒ Use Data Interpreter
Data Interpreter might be able to connect your data sources faster.

APAC COGS

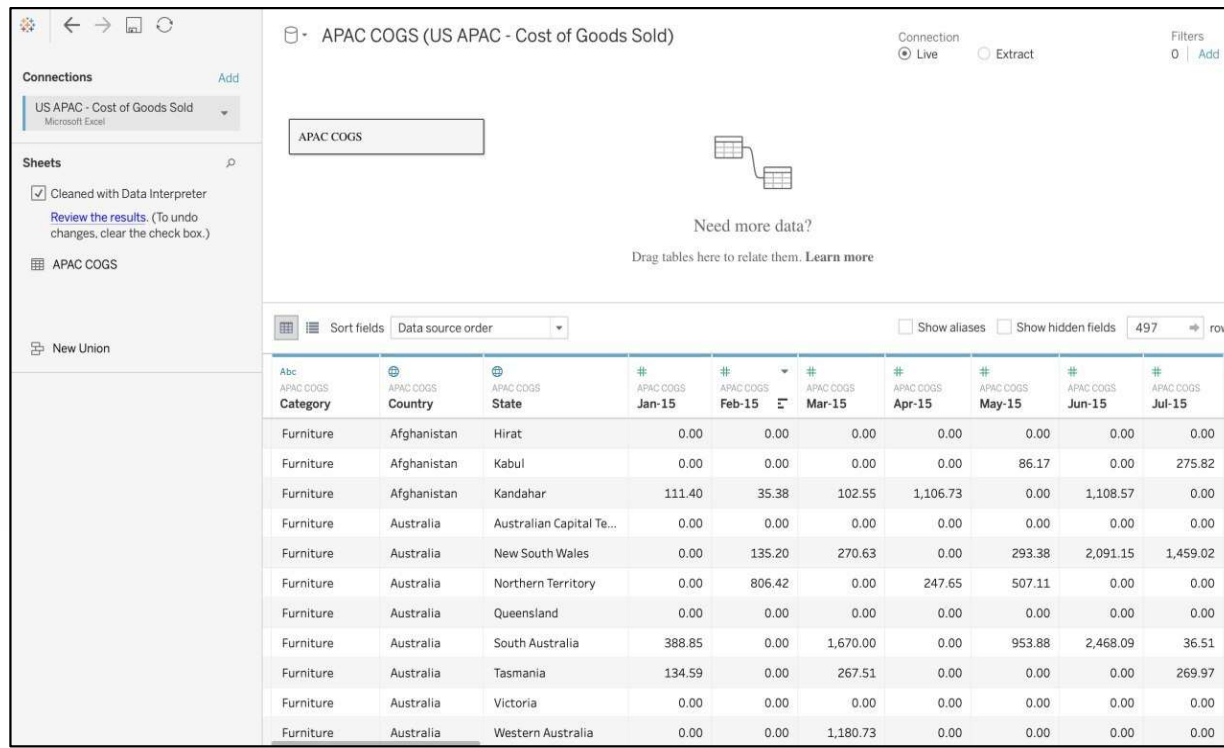
New Union

Need more data?
Drag tables here to relate them. Learn more.

APAC COGS	APAC COGS	APAC COGS	APAC COGS	APAC COGS	APAC COGS	APAC COGS	APAC COGS	APAC COGS	APAC COGS	APAC COGS	APAC COGS
F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	
Superstore Cost of S...	Category	State	42,005.00	42,036.00	42,064.00	42,095.00	42,125.00	42,156.00	42,186.00	42,217.00	
Furniture	Afghanistan	Kabul	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
		Kandahar	0.00	0.00	0.00	0.00	86.17	0.00	275.82	0.00	
		Kandahar	111.40	35.38	102.55	1,106.73	0.00	1,108.57	0.00	1,592.1	
	Australia	Australian Capital Te...	0.00	0.00	3.00	0.00	0.00	0.00	0.00	0.00	
		New South Wales	0.00	135.20	270.63	0.00	293.38	2,091.15	1,459.02	2,994.8	
		Northern Territory	0.00	806.42	0.00	247.65	507.11	0.00	0.00	0.00	
		Queensland	0.00	0.00	3.00	0.00	0.00	0.00	0.00	913.0	
		South Australia	388.85	0.00	1,670.00	0.00	953.88	2,468.09	36.51	492.6	
		Tasmania	134.58	0.00	267.51	0.00	0.00	0.00	269.97	1,264.5	

Blend (cont'd)

5. Based on the view, we can see that there is (a) *null* in some of the cells and, (b) no appropriate column name. This data requires data cleansing. In Tableau, we can use (c) - **Use Data Interpreter** to fix this problem. After you tick on the button, your data will be look like as shown below.



Category	Country	State	Jan-15	Feb-15	Mar-15	Apr-15	May-15	Jun-15	Jul-15
Furniture	Afghanistan	Hirat	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Furniture	Afghanistan	Kabul	0.00	0.00	0.00	0.00	86.17	0.00	275.82
Furniture	Afghanistan	Kandahar	111.40	35.38	102.55	1,106.73	0.00	1,108.57	0.00
Furniture	Australia	Australian Capital Te...	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Furniture	Australia	New South Wales	0.00	135.20	270.63	0.00	293.38	2,091.15	1,459.02
Furniture	Australia	Northern Territory	0.00	806.42	0.00	247.65	507.11	0.00	0.00
Furniture	Australia	Queensland	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Furniture	Australia	South Australia	388.85	0.00	1,670.00	0.00	953.88	2,468.09	36.51
Furniture	Australia	Tasmania	134.59	0.00	267.51	0.00	0.00	0.00	269.97
Furniture	Australia	Victoria	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Furniture	Australia	Western Australia	0.00	0.00	1,180.73	0.00	0.00	0.00	0.00

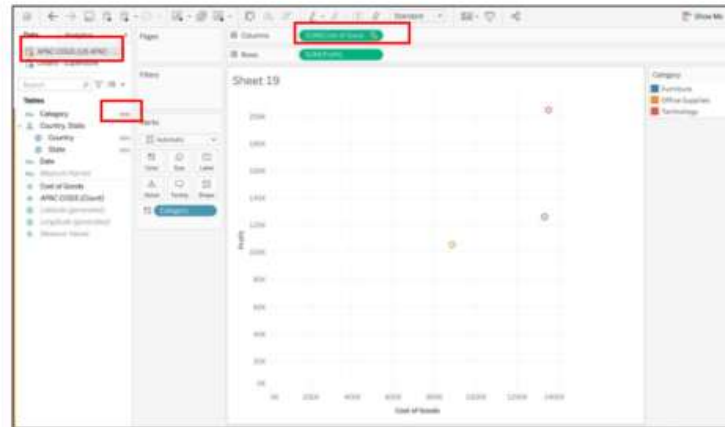
Blend (cont'd)

6. Now, the problem with null data has been solved. However, the cost values are separated by date and the date have become column names. We need to fix this by changing it into a single column instead of rows.
7. Click or  (**Manage metadata** icon). Then, click on **Jan-15** column. On your keyboard, hit and hold the **Shift** button, and click on **Dec-18** column.
8. Right-click on the highlighted area, and select **Pivot**.
9. After pivoting, the highlighted columns will be transformed into two columns only. Rename Pivot Field Names as **Date**, and Pivot Field Values as **Cost of Goods**.

Field Name	Table	Remote Field Name
 Category	APAC COGS	Category
 Country	APAC COGS	Country
 State	APAC COGS	State
 Date	Pivot	Pivot Field Names
 Cost of Goods	Pivot	Pivot Field Values

Blend (cont'd)

- Go back to preview data source by clicking on the icon 
- Next, go back to the most recent sheet. Look on the Data tab, you can see there are two different data sources. The one with blue tick is the primary data source for the analysis.
- In the Data pane (under APAC COGS), drag **Cost of Goods** to **Columns** shelf.



You will see there is an orange tick at APAC COGS data source, which indicate this is a secondary data source in this analysis.

Both of these data sources are linked by one unique key. In this case, Tableau detect both data source contains Category field, hence it automatically creates link using this field.

Blend (cont'd)

13. Change back into Orders – Superstore data source. Drag **State** into center of the canvas.
14. Do some adjustment by changing the mark type to **Circle**. Then, alter the color by reducing the opacity and select black color for border.
15. Rename this sheet as “Cost vs Profit”.

Exercise

- 1) Which month does the company has the most average sales?
- 2) Which region has all positive profit in all sub-category?